

The Real State: Inside the Congo's Traffic Police Agency[†]

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This paper provides insight into a corruption scheme in Kinshasa's traffic police agency. First, various data collection branches show that the agency's revenue is five times that from fines and is derived from a coalition of traffic police officials, their managers, and judicial police officers scheming to extort drivers. Second, the analysis of an experiment suggests that the scheme subverts service. Third, the scheme appears to be a rational response to the context, but its logic is widespread. The findings suggest that coalitions of officials, while being socially costly, can yield large illicit revenue, nuancing the notion of state weakness. (JEL D73, H76, K42, O17)

A large literature in economics has examined the causes and consequences of corruption. Much of that literature posits corruption as the problem of an individual state official with discretion to use office for personal gain. Their incentives can be

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misaligned with the functions of the state and thus be costly for society (Shleifer and Vishny 1993; Bardhan 1997; Olken and Pande 2012).

However, state officials are part of state organizations, hence rarely act in isolation. Indeed, managers often know about corruption by their agents and even take part in it; sometimes, their involvement might enable schemes that would otherwise be infeasible by individual officers. When officials form coalitions with their own managers, corruption could become invisible to the naked eye, and both its revenue and social cost could be much larger. In a seminal study, Wade (1982) describes one of such schemes in irrigation canals in India and suggests that the fact that corruption is a scheme may further exacerbate its social cost. Yet research about these schemes is based on that one remarkable but qualitative study. The reason for this gap is obvious: "Given the extreme sensitivities at stake ... one cannot work towards an understanding of the phenomena discussed here by the familiar methods of the random sample, the formal interview, and the structured questionnaire. One has to use, rather, more informal, more anthropological means" (Wade 1982, p. 291).

Attempting to fill this gap, this paper reports on a study that builds upon trust relationships developed during three years of qualitative fieldwork inside one specific agency: the traffic police commissariat of Kinshasa, Democratic Republic of the Congo (DRC). Drawing on these relationships, we experimentally varied in 2015 the managers' corruption schemes with their agents and designed multiple independent data-gathering activities. Complementing the information gathered through these activities with three additional field visits and nine years of qualitative interviews thereafter, we detail how the corruption scheme operates in the commissariat, *quantify* the magnitude and nature of its profits, and *estimate* its effect on public service.

Kinshasa's traffic police commissariat was well suited to answer those questions. Like many traffic police agencies worldwide, officially, the dual mandate of the agency was to manage traffic and to issue charges in order to deter drivers from violating the traffic code. Police commanders (henceforth, managers), each posted at one of Kinshasa's 29 traffic police stations, deployed officers (henceforth, agents) in teams to the intersections under their jurisdiction. The agents' official mandate was to manage traffic and to escort drivers who violated the code to the stations, alongside producing allegations of infractions made by drivers, so that the station officers could charge the drivers. Unofficially, agents were regularly seen taking bribes from drivers in the street. In the backdrop of this, managers purportedly asked the agents to escort drivers daily to the station (Baaz and Olsson 2011), where station officers purportedly took bribes from the drivers (station bribes) in exchange for not charging the driver for alleged infractions that were sometimes reportedly arbitrary (i.e., extortion). Relationships built over three years provided an opportunity to analyze this scheme, locally known as the 'quota scheme.'

The first key input into our analysis is a randomized encouragement, designed in 2015 during 30 days inside one of the two battalions of Kinshasa commissariat, including 7 of the battalion's 14 stations, as well as Kinshasa's central station, thus totaling 8 stations (out of 29 in the city), 8 station managers, 16 judicial police officers, 16 fine collection agents (henceforth, FCAs), and 15 teams of agents, consisting of 63 agents. We randomly assigned, independently for each team, 15 of the team's 30 days to an encouragement given to the corresponding team's manager for the team's quota not to surpass a cap that day (henceforth, quota reduction encouragement). Each

cap was half of the team's average quota level. The managers were compensated for the loss due to the reduction and knew that some days and teams would sometimes receive a quota cap encouragement, but they did not know when nor that its effect in the street was measured. Accounting for festivities and attrition, the sample (henceforth, experiment sample) is an imbalanced panel of 337 team-days.

The second key input into our analysis is a dataset (Sanchez de la Sierra et al. 2024) on the universe of transactions in the experiment sample and the preceding day (henceforth, Day Zero). To construct these data, we designed five *independent* branches after collaborating with police colonels, station managers, judicial police officers, the Fine Collection Agency (DGRAD), the Transportation Division, National Drivers' Association members, and civilians. Overall, 160 individuals took a direct part in data collection activities. These branches relied on direct observation, self-reported data, and administrative data. We ensured that none of the individuals whose publicly observable behavior was recorded by a third party knew that their behavior was being recorded, with meticulous tailoring to community norms and ethics (see the Ethics Appendix, shown in online Appendix, p. 68). The data from this endeavor include records of (at least) 1,255 vehicles inside the stations from 2 independent sources and verified by administrative records, 13,793 unique police agent/driver interactions in the street, 6,399 records of 2,997 hourly traffic outcomes, and 337 team-day vehicle quotas constructed from 7 sources.

Our first result is descriptive. Using the union of Day Zero and the experiment sample's team-days not assigned to encouragement (henceforth, descriptive sample), and verifying the validity of this sample for descriptive analysis, we document the following three facts about the scheme.

Fact 1: The quota scheme generated the battalion's *real*, if mostly unofficial, revenue. Overall, driven by the station bribes, the real revenue was 5.2 times larger than the revenue from fines.

Fact 2: At the stations, station officials routinely took station bribes from the escorted drivers, seemingly whether or not the driver had made an infraction; that is, *extortion*. Station bribes were determined through an informal negotiation between the escorted driver and the station officials, in exchange for the station officials not charging the driver with an infraction; alleged infractions were often (67 percent of times) not verifiable by a third party (henceforth, unverifiable). Drivers who were unable to call a high-ranked state official to advocate for them during this negotiation (commonly called a "protector") paid 2.2 times more in station bribes than those who could.

Fact 3: Negotiation over station bribes was facilitated by the agents' apparent ability to arbitrarily escort drivers to the station using force and make discretionary allegations of infractions. Qualitative notes indicate that agents often brought "fantasy" allegations to the station. Quantitatively, in 82 percent of pull-overs, observers reported that no infraction was visible in the street. In those, disputes or escapes were much more likely than those in which they reported one.

Our second result is that the quota scheme undermines public service. We use the experiment to analyze the effect of the quota on both of its mandates. *In the first*

step, we analyze traffic outcomes. We find that the quota accounts for a large share of accidents and traffic jams at the intersections. This effect is partly (but not entirely) channeled by agent absenteeism induced by the effort to escort drivers. *In the second step, we analyze deterrence.* We provide evidence suggesting that the quota waters down the benefits faced by the drivers of respecting the code. Guided by a simple multitasking framework with a manager having the power to issue wrongful charges, we examine two testable implications for whether the quota improves or undermines deterrence. First, if the power to issue wrongful charges on escorted drivers is large—that is, extortion, a necessary but not sufficient condition for the quota to undermine deterrence—the quota should induce agents to escort drivers who, irrespective of whether they committed an infraction, can be predicted to yield a large station bribe. Consistent with extortion, we find that the quota only increases the escorting of drivers without a protector and of those whose predicted station bribe, based on their characteristics, is high—against those with a high predicted fine. Second, the quota undermines the incentives of drivers to comply with the code if, and only if, it produces more false than true allegations. We find that the quota leads to more escorts of drivers with unverifiable alleged infractions and pull-overs of drivers in which the driver is the least likely to have made a true infraction. Overall, this suggests that the quota weakens both traffic management *and* deterrence.

Our third result is that the quota scheme can be rationalized as an equilibrium corruption contract in this setting. First, bargaining between the agents and the drivers to avoid costly escorting often breaks down because some drivers are liquidity constrained, are often agents for another person, and because of mutual *spite* ensuing often tense pull-over interactions; this explains why the quota does not unravel in equilibrium. Second, the level of the quota does not affect fine revenue, suggesting it is about corruption. Third, reducing the quota reduces total corruption revenue, suggesting that it might be the first best for the manager-agent pair.

We then place these results in perspective. First, we show, based on our qualitative interviews, that while a scheme has existed since the 1980s, its specific form as a vehicle quota developed when local stations were created in 2010, became entrenched after a crackdown on street-level bribes, and remained thereafter because managers found it more profitable than asking for a fee. Second, describing similar schemes across the world, we document that while the way in which a scheme is organized is context specific, its economic logic appears to be quite general.

These findings make three contributions. First, they document the role of manager-agent schemes in corruption. Our study is not the first to measure corruption. Indeed, a rich literature in economics has developed ingenious ways to measure corruption's prevalence, causes, and consequences (Olken and Pande 2012; Banerjee, Hanna, and Mullainathan 2013).¹ Our study does not introduce the notion

¹Through these techniques, officials in Uganda were shown to price discriminate in the setting of bribes paid by firms (Svensson 2003); state officials in India were shown to misallocate driving licenses (Bertrand et al. 2007); state officials along the roads in Indonesia were shown to create sophisticated pricing schemes for their bribes designed to extract the drivers' willingness to pay, leading to suboptimal incentives (Olken and Barron 2009); and coercive bribe taking in South African ports was shown to be costly trade diversion (Sequeira and Djankov 2014). Other studies collected prices paid for services as a measure of corruption (Banerjee, Deaton, and Duflo 2004), which show that bribes induce misallocation of health services. Another ingenious strategy has been cross-checking the reports of a particular outcome with an independently collected measure. Reinikka and Svensson (2005); Fisman and Wei (2004); and Olken (2007) provide remarkable examples of the promise of this approach.

that managers may also be corruptible. Indeed, it is a known problem in a large literature on the economic theory of organizations (Tirole 1986; Dixit 2012). Our study is also not the first to *empirically* document manager participation in corruption; in a remarkable qualitative study, Wade (1982) documents a sophisticated system in which the engineers in a canal irrigation agency redistributed revenue to superior officers and politicians, showing that corruption is connected through the organization. Mutahi, Micheni, and Lake (2022) and Baaz and Olsson (2011) document qualitative evidence of similar practices in the police in Kenya and in Kinshasa. By leveraging trust relations to design an experiment on a scheme itself, our study presents the first causal evidence of the effect of corruption schemes; by scaling up methods from qualitative research into multiple independent data collection systems, it offers the first dataset on *one* corruption scheme. Our findings' contribution is that manager schemes can add huge profits and *distort* corruption's social costs beyond corruption without such schemes.

Second, these findings document extortion by state officials. Our study is not the first to analyze extortion. There is a rich literature in economic theory on the optimal incentives for state officials in the presence of bribery and extortion (e.g., Polinsky and Shavell 2001; Khalil, Lawarrée, and Yun 2010; Angelucci and Russo 2016). A consensus is that extortion is costlier than bribery because it disincentivizes citizen compliance. Our study is also not the first to document extortion: Olken and Barron (2009) document the military on Indonesian roads carrying guns and asking for a bribe, suggesting they have, and use, the power to extort drivers. Our study's contribution is to present evidence characteristic of extortion, based on the types of vehicles stopped and the veracity of allegations, as well as the role of the managers in extortion.

Third, a growing literature in economics analyzes state effectiveness (e.g., Weigel 2020; Bergeron, Tourek, and Weigel 2021). Much of that literature has examined engineering of the state's official organization to promote its official prerogatives, such as taxation. Our findings reveal that the total revenue of a state agency is five times its official revenue. Regardless of whether it is illegal, this constitutes real revenue-generating capacity. This discrepancy between the official and the *real* state functions resonates with a literature in anthropology, which argues that "real governance" is central to what states do (Migdal and Schlichte 2005; Titeca and De Herdt 2011; Lund 2006; Olivier De Sardan and Herdt 2015).

I. Kinshasa's Commissariat of the DRC's Traffic Police Agency

Aiming to contain congestion and accidents, the traffic police agency's mandate is to manage traffic flows and to charge traffic code violators. Its goal is valuable: the DRC's road traffic death rate was among the world's 10 highest in 2022 (World Health Organization 2022). In 2015, the traffic police commissariat of Kinshasa, a city of 17.2 million, was divided into 2 battalions, each with 14 stations, alongside Kinshasa's central station. Each station was run by a commander (henceforth, the manager), assisted by 10 officers in the street (henceforth, agents) and 2 judicial police officers in the station (henceforth, JPOs), who worked for the Office of Public Prosecution.

Officially, the manager had the power to authorize his agents to work in teams at street intersections (henceforth, intersections). One of the agents, the team leader,

monitored the team for the manager. The agents' official power mirrored the agency's two mandates. For traffic management, they could order drivers where to go; refusing an order was a code violation (in French, *refus d'obtempérer*). For charging violations, they could escort violators to the station.² The station was much like a tribunal: the JPOs acted as a judge (i.e., they had the power and discretion to charge drivers), the driver as the accused, the agent as the witness/prosecutor (i.e., they had the power to make reports of alleged infractions). The FCAs would issue a fine collection note for charged drivers to pay at the bank; the receipt thereof would entitle the driver to retrieve the vehicle. The agents and station officers' official salary is \$69 and \$100 monthly, respectively.

Unofficially, the managers were said to request their agents to bring a previously agreed daily target of vehicles to their station (the vehicle quota scheme). At the station, where JPOs were often idle waiting for drivers, drivers purportedly paid station bribes of two types. First, they reportedly paid the JPOs a bribe to avoid being charged (henceforth, charge negotiation bribe). Since the JPOs threatened to charge the driver based on an alleged infraction produced by the agent, station bribes amounted to collusion between the JPOs and the drivers—bargaining to avoid a charge in the presence of evidence of infraction. However, the allegations made by the agents could reportedly be arbitrarily false; thus, under the cover of collusion between the driver and the JPO, station bribes could sometimes amount to the station officials colluding with the agents in order to extort drivers. Second, the drivers also had to pay an unofficial pound fee to recover their vehicle (*frais de fourrière* in French, henceforth, car pound bribe), which was often less significant than charge negotiation bribes. JPOs reportedly shared station bribes with the managers, who then purportedly paid higher-ups in the agency's hierarchy. Upon meeting their quota, the agents focused on collecting informal toll fees (the “salute of the cola nut,” *mbote ya likasu*, in Lingala) and occasional voluntary contributions from drivers and managed traffic; rarely, they made unofficial payments to their managers (the *retrocession* in French). The other actor in the stations, the FCA, had little discretion. While fine payments at the bank were official from the perspective of the driver, it is possible that some may not have reached government coffers. However, prior to the study, we did not find anecdotes supporting that FCAs or bank agents took bribes.

This scheme relied on unofficial sources of power but also on power officially bestowed upon the officials. First, as a pillar of the vehicle quota, managers used their officially bestowed power over the agents' jackets, which were required for the agents to work in the street, to threaten to suspend them or to deploy them to places where they would make little bribe income for themselves (so-called sterile intersections). Also, managers' kickbacks to higher-ups purportedly cultivated an “umbrella” that, according to our interviews, allowed them to “better exploit their agents.” A manager without an umbrella trying to impose a quota on his agents could risk being accused by his agents to higher-ups and thus risked official and unofficial sanctions. Second, to obtain station bribes from the drivers, the JPOs used the official power bestowed upon them by the Office of Public Prosecution, giving them the discretion to charge drivers. Informal negotiation between the JPO and the

²Online Appendix Figure A1, panels A–D show the hierarchy, the stations, a map of Kinshasa, and a roundabout, respectively.

driver, who often called someone (the “protectors”) to talk to the JPO on their behalf, was reported to be central in the determination of the station bribes; this purportedly could include intimidation. Third, the agents used the power of discretion officially bestowed upon them to pull over and escort drivers to the station. They were sometimes reported to have used that to produce false allegations in service of the JPO’s negotiation with the driver at the station; in that case drivers called the interaction with the agents the *tracasserie* and perceived it as illegitimate (Baaz and Olsson 2011). The agents used various tactics, in some cases psychological or physical intimidation, coercion, humor, or violence (e.g., online Appendix Figure A1, panel E). It was common to see agents pinch tires, enter vehicles, confiscate keys, licenses, or radios, as one driver lamented in a press article: “Officers brutalize us, tear out our license plates. Too many *tracasseries*. After having seen our documents, [they] invent imaginary infractions” (Radio Okapi 2021).

Attempts to escort a driver did not always result in an escort to the station; indeed, the driver could escape, or, in rare cases, the agents agreed to let the driver go, foregoing being escorted in exchange for a bribe paid to the agent in the street, called the *amende* in French.

The officials above the managers considered the scheme to be a self-financing tool for the police and were qualitatively described as tacitly encouraging it, as it purportedly allowed them to pocket the funds that were originally justified to higher levels to cover the stations’ costs; furthermore, it purportedly generated kickbacks for them too.³ On the other hand, it was also perceived to undermine service. First, since agents purportedly could fabricate allegations, few drivers in Kinshasa met basic requirements, investing instead in creating connections with powerful officials.⁴ As a result, the scheme was perceived to reduce safety. Second, in a context with no traffic lights, *tracasseries* purportedly distracted the officers from managing traffic, leading to disorder. The rest of this paper analyzes this scheme and its effect on public service.

II. Experiment Design: Encouragement to Temporarily Reduce the Quota

The experiment was designed in 2015 in 1 of the 2 battalions of Kinshasa’s commissariat, covering 7 of its 14 stations as well as Kinshasa’s central station. This thus covered the staff of eight police stations. Such staff was composed of 8 managers, 16 JPOs, 16 FCAs, and 15 teams of agents permanently posted to 15 distinct intersections, totaling 63 agents (13 teams had 4 agents, one of 5, and one of 6). We now describe the experiment design.

A. The Encouragement to Reduce a Team’s Quota for a Day

The experiment was a clustered randomized encouragement design, where an encouragement was independently given at the level of team-day clusters to temporarily reduce the quota for one day. Specifically, the encouragement at a given

³The managers purportedly drew on station bribes revenue to cover the gas expenses (Nkuku 2022).

⁴The following encounter, described by various observers, illustrates the importance of protection relationships: “The agent and a soldier who owned a motorbike quarreled over the motorbike. The soldier said, ‘tomorrow you’ll see what I do; I will come back!’ The soldier came today with a jeep, heavily armed. They arrested the agent. They threw him in the jeep in front of everyone! The soldiers left with the agent, and they are going to beat him up.”

team-day involved a previously vetted member of the research team asking the team's respective manager, on a given team day, to request no more than a capped number of vehicles from that team that day to the station through the quota.

The design leveraged the following features of the context. First, while quotas varied across team-days, *each team's* modal quota was known to the participants (henceforth, natural quota). The cap was always half the team's natural quota. We obtained the natural quotas in six months of qualitative fieldwork prior to the study; the encouragement was worded as not requesting more than $[X]$ vehicles that day from this team, where X had previously been calculated to be half the natural quota. Second, the managers communicated the quota to each team in a routine early morning phone call. Thus, the research team called the team's manager each day *prior to this call* and communicated which of his teams had been selected for the cap that day alongside the cap of each. We reminded managers, including those for whom no team had been selected that day, that teams not assigned to the encouragement could have any quota. Third, each team was permanently assigned an intersection; thus, we expected the encouragement to affect the outcomes of that intersection. Fourth, the quota was the main source of manager income; hence, managers were compensated for the estimated loss in income caused by the caps.

Implementation: First, the managers were informed in private that, for approximately one month, for some days, the research team would ask them not to surpass a quota of some teams and that they would be compensated for the loss. The managers did not know how frequently each team would be selected nor what days were selected. Second, the authors only informed the research team of the daily cap assignments at the start of each day. Third, as it was their main source of revenue, we compensated the managers for the loss in their revenue induced by our reduction. To calculate the compensation, we gathered the average manager bribe revenue per vehicle in preparatory fieldwork *before* informing managers about a reduction. Fourth, to avoid influencing operations, those who administered the compensation (henceforth, encouragement team) were from police networks, with access inside the stations authorized by the managers; they compensated the managers privately early in the morning. Fifth, the managers were informed that if they took the compensation but failed to reduce the quota, they would forgo participation in the project. An auditor, flown from another province to prevent collusion with the encouragement team, verified daily whether the managers had received the assigned compensation. Sixth, while weekday, month, calendar day, assignment lag, pre-study station bribe income, and number of agents per team are balanced across assignment status in the experiment sample, there is a mild imbalance in the encouragement assignment across calendar days, which we account for in the analysis of the experiment (online Appendix Table A1).

Ethics: Compensating the managers for their quotas' reductions can raise the ethical concern that this may feed into a corrupt system. Key features of the context shed light on this concern. First, the quota is a corruption scheme. Thus, the encouragement amounts to reducing corruption by compensating the officials benefiting from corruption for the loss of their main source of income it induces. As such, it extends existing anti-corruption interventions aiming to examine whether financial incentives can reduce corruption, targeting the managers instead and further strengthening the incentive by conditioning it on reducing corruption. Second, the compensated

encouragement, obtained legally, substitutes the illicit manager income obtained through extortion of drivers for questionable infractions and based on mutual threats. It remains possible, however, that engaging with commanders on an existing quota level amounted to a recognition of the legitimacy of the system. However, from the perspective of the scheme's participants, the approval of foreign researchers or of students is not relevant to determining whether the scheme is legitimate. Since the 1980s, a huge number of actors have formed a part of, benefited from, and been aware of the scheme. Their participation and the reception of society at large, more than the view of foreign researchers determines whether they perceive it to be legitimate. Ascribing the power to legitimize the system to these researchers, who are not working with the government, would imply misrepresenting the system, its relation to the social context, and the researchers' relevance to these actors. The Ethics Appendix (shown in online Appendix, p. 68) discusses the ethics of this study comprehensively.

B. From Random Assignment in the Randomization Sample to the Experiment Sample

The randomization sample is 600 team-days, spanned by 15 teams over 50 days. Independently for each of the 15 teams, we randomly assigned the encouragement to 25 out of the team's 50 days in the randomization sample (the randomization clusters); thus, the probability of assignment was equal across teams. This ensured that 300 team-days were assigned to encouragement and 300 to no encouragement. The randomization sample was chosen under uncertainty over the budget; thus, the stopping rule was determined by budgetary constraints. Thus, to further reduce the likelihood of an imbalance, randomization was stratified within Team $\times$ Day 1–25 versus Team $\times$ Day 26–50 (the 30 randomization strata).

This stopping rule pinned the end of the experiment on day 31. The proper subset of the randomization sample in which we aimed to conduct the experiment (henceforth, target experiment sample) was thus constituted of 465 team-days, which preserved their random assignment: 241 were assigned to the encouragement, and 224 were not. Then, in the target experiment sample, closure due to festivities and the removal of two teams for precaution led to the loss of 90 and 38 team-days, respectively (44 and 17 of which were assigned to the encouragement, respectively), resulting in a sample of 337 team-days (the experiment sample).⁵ Figure 1 presents a Consort diagram (Schulz, Altman, and Moher 2010).⁶

III. Data Collection and Data Construction

In this section, we describe the sample, data collection, and data construction. The data and source code are in Sánchez de la Sierra et al. (2024).

⁵First, there were 5 Sundays and 1 festivity, leading to the loss of 6 days for each of the 15 nondropped teams—that is, 90 team-days. Second, in each of days 3, 5, and 6, we replaced 1 team, leading to no loss; the teams had been reshuffled. On day 8, we dropped 2 teams, leading to the loss of 38 (nonfestivity) team-days; we had learned that the agents were tense and feared spying by the general (remnants of the “Kanyama psychosis”; see Section VIIA).

⁶Online Appendix Figures A2 and A3 and online Appendix Table A2, panel A present the assignment graphically, in a map, and the number of teams in each station-day. A reporting checklist is in online Study Documentation, Section 1.

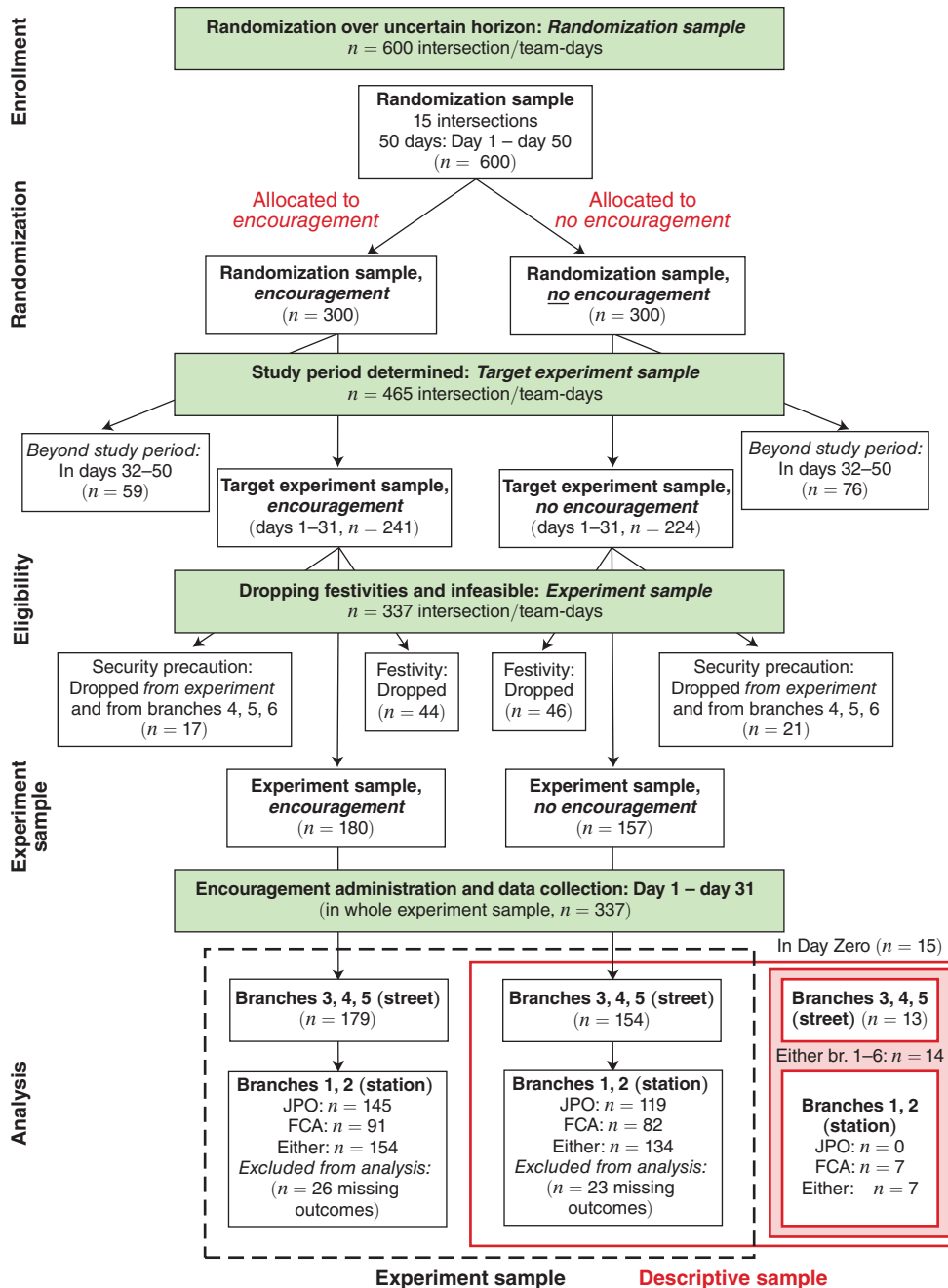


FIGURE 1. CONSORT DIAGRAM: ASSIGNMENT TO THE ENCOURAGEMENT AND SAMPLES

Notes: This figure presents the Consort diagram (Schulz, Altman, and Moher 2010) representing the assignment to the encouragement and the samples. We use Consort reporting guidelines through the paper (Schulz, Altman, and Moher 2010). We report the checklist for randomized trial in the Study Documentation (shown in online Appendix, p. 48).

A. From the Experiment Sample to the Data Collection Sample

We designed 5 data collection branches inside the 8 stations and 15 intersections (with 63 unique agents) of the target experiment sample to cover the 30 days of the target experiment sample and Day Zero. As described in Section IIB, the target experiment sample was afflicted by festivities, attrition, and replacements; those affected the data collection activities in this sample, too. Furthermore, on Day Zero, some branches could only cover 14 teams. Thus, the branches were implemented in the 337 team-days of the experiment sample and the 14 teams on Day Zero. This produced an imbalanced panel of $337 + 14 = 351$ team-days (henceforth, data collection sample), composed of 8 stations and their 8 managers, 16 JPOs, 16 FCAs, and 63 agents (and 12 replacements) in 15 intersections (and 3 replacements), shown in online Appendix Table A2, column 1.

B. The Five Data Collection Branches

Branch 1: Data Reported by Judicial Police Officers inside the Stations

Context: The JPO had firsthand knowledge of the intersection of origin, time of arrival, vehicle's make, brand, and function of the vehicle. The JPO made the initial request to the driver and had a private conversation with protectors, due to the call being sensitive, and then decided whether to charge the driver and whether to receive station bribes, often within the day. Thus, the JPO had firsthand knowledge of the alleged infraction, initial bribe request, protector, station bribes paid, and whether the JPO charged the driver.

Collection: In May 2015, we obtained the approval of the colonel in charge and from each station commander for the following data design. The JPOs of each of the eight stations recorded, on paper, for each escorted vehicle, the following information: intersection of origin and time of arrival, make and brand, vehicle's type, allegation (open-ended), initial bribe request, employer and rank of the driver's protector, station bribes, whether they were charged, and the requested fine amount if charged. The JPOs also included each team's quota level daily. They received minimal compensation for their time. This produced details about 1,235 driver negotiations in the stations from the data collection sample, of which 95 were from unidentified intersections (the "Unassigned intersections sample"), respectively, likely from the data collection sample (online Appendix Table A2, panel B). Furthermore, in each station, the JPO data collection supervisors corrected the JPO reports using already existing bribe records.⁷ The JPO data collection supervisors also had informal discussions in the station daily and recorded quota levels independently (Branch 1b).

⁷"The JPOs have a register in which they keep the cars for the FCA agents; but they also have a single paper on which they have written the true entries. This register is for the commander, to show what is sent to the FCA agent and what is for the police. We cannot copy this register, but we use it to verify the JPO data. Since about the 27th of June, we access it daily." Source: Evening data collection supervisors meeting, June 2015.

Branch 2: Data Reported by the Fine Collection Agents inside the Stations

Context: The FCAs sat in the same station as the JPOs, which was typically a single-office small container. FCAs were routinely aware of station bribes, which were discussed publicly in the office because those were not sensitive. However, without exerting additional effort, they did not easily know the brand and make of the car, the intersection and hour of origin, whether there was a protector and their identity, nor the allegation for drivers that were not charged.

Collection: The fine collection agency agreed on the following design in May 2015. The FCAs of each of the eight stations recorded the station bribes of each escorted driver based on hearing the conversation. They also reported their guesses about the intersection of origin, time of arrival, allegation, brand, vehicle type, daily quota level for each team, and, for each charged driver, the fine paid. The FCA forms were blank to prevent Hawthorne effects. The FCAs received minimal compensation for their time. Reflecting their poor knowledge of the intersection of origin, Branch 2 produced details of 728 and 527 negotiations classified as from the data collection sample or from unassigned intersections, respectively (online Appendix Table A2). This design ensured that Branches 1 and 2 were independent.⁸ In addition, the Branch 2 supervisors also had informal discussions in the station each day and recorded quota levels independently (Branch 2b).

Branch 3: Data Reported by Street Observers from the Street Intersections

Agents were required to wear an orange jacket and a yellow helmet with large numeric identifiers; thus, they were easy to identify. The Division of Transports of Kinshasa agreed on the following design in May 2015. In each of the 15 intersections, 2 observers on opposite sides of each intersection recorded the events of agents interacting with the drivers in a smartphone app. Each observer recorded the universe of agent-driver interactions (events) of one randomly selected agent (henceforth, target), based on a roster of agents of each station obtained prior to the study. For each event, they recorded the bribe type and amount, whether it was a pull-over, an escort, the vehicle type and brand, and described the interaction in open-ended format. This produced details about 13,793 police-agent/driver interactions in the data collection sample (online Appendix Table A2, panel B), of which 1,841 were pull-overs. In addition, each round hour, observers entered whether, in the preceding hour, there was a traffic jam or an accident and, during one minute, estimated the number of passing vehicles and their average speed. This produces 6,399 measurements about 2,997 day/hour-intersection traffic outcomes.

To ensure their safety and to minimize Hawthorne effects, observers were all employees of the Division of Transports. They worked alongside the beverage, phone credit, or currency exchange retailers that routinely operate at intersections; they gathered naturalistic observations of publicly observable behavior on which there

⁸Based on a large amount of information obtained daily from the stations, including suspicions and conflicts, we were convinced that FCAs and JPOs were never aware of the existence of Branch 1 or Branch 2, respectively.

could be no expectation of privacy—thus, the agents did not need to be informed—and we randomly assigned them each day to different intersections.

Branch 4: Quota Level and Retrocession Internally Reported by the Agents

After hours, the agents wrote a short diary about their day on a blank page and received appropriate compensation to complete it. (See the Study Documentation shown in online Appendix, p. 48.) Diaries were collected after hours by two local researchers considered internal to police networks who had no knowledge of the rest of the study (henceforth, “assistants”). In the diaries, the agents reported information about their daily consumption, retrocession, and the team’s quota that day.

Branch 5: Quota Level Data Verified by External Auditor from the Agents

The auditor also informally recorded the teams’ quotas once personal trust had been built. His role was introduced as an informal conversation verifying that the assistants had provided the diary compensation. That he was flown from another region reduced the risk of collusion with the agents or the assistants.

Online Appendix Figure A4 and Table A3 visualize the branches for a typical station and summarize the data. The Ethics Appendix (shown in online Appendix, p. 68), Section III, and the Study Documentation (shown in online Appendix, p. 48), Section II, describe data collection’s ethics and management, respectively.

C. Data Construction

For station activity, Branch 1 was designed to measure more outcomes, whereas Branch 2 observers had better incentives to report station revenues.⁹ Thus, we use Branch 1 in all analyses and complement it with Branch 2 for station revenues. Since infractions are made in the street, no station branch includes information on whether the alleged infraction is true; hence, we manually classify the open-ended descriptions of the alleged infractions in Branch 1 into whether it could be verifiable by a third party (e.g., not *owning* a driver’s license) or not (henceforth, unverifiable). *For agent behavior*, we construct a team-day aggregate (Study Documentation (shown in online Appendix, p. 48), Section III). Branch 3 recorded 13,793 events; of those, 1,841 were traffic pull-overs. Branch 3 reported escorts are highly correlated with station reports of escorts, but Branch 3 also includes pull-overs that did not result in escorts; thus, we use Branch 3 to analyze pull-overs.¹⁰ We use Branch 3 open-ended descriptions to classify street-level events into whether the agent used force, whether

⁹Online Appendix Table A4, panel A shows that the correlation between Branches 1 and 2 for the number of escorts and revenue (columns 1–6) is high and significant. This provides confidence in each independent source’s quality to measure station revenues. Second, the mean number of escorts and mean bribe and fine revenue are larger in Branch 2. This is consistent with Branch 2 yielding better information about station bribes revenues. Third, columns 7–16 support that only Branch 1 has usable information about protectors and allegations; there, the correlations are always nonexistent, and Branch 1 means are much larger.

¹⁰Online Appendix Table A4, panel B presents estimates of a regression of Branches 1 and 2 on Branch 3 escorts (and vice versa). First, mean escorts of an intersection-day are similar across branches; unsurprisingly, they are larger for Branch 3 since some of those were not to the station. Second, the correlation is strong and significant; given the independence of the sources, this provides reassurance that they are informative.

there was a dispute, and, for traffic pull-overs, whether the driver clearly made an infraction. *For traffic outcomes*, we construct day/hour-intersection measures of traffic outcomes as the mean of each intersection's two observers' reports of each day/hour-intersection traffic conditions. Simple checks provide reassurance in the reliability of these data.¹¹ *For quota levels*, we use the mode of the team-day quota levels reported in Branches 1, 1b, 2, 2b, 4, 5, and of the outcome of daily research staff meetings where quota levels are deliberated.

IV. The Real Revenues of the Traffic Police Agency

Drawing on the data gathered through these five data collection branches, we now establish three facts about the scheme and its significance for corruption.

A. From the Data Collection Sample to the Descriptive Sample

For some analysis in this section, we create a random sample of team-days from the data collection sample not assigned to the encouragement (henceforth, descriptive sample). Specifically, it is the union of the 14 team-days in data collection sample's Day Zero and the 157 team-days from the experiment randomly sampled to *no* encouragement, totaling 171 team-days (online Appendix Table A2, panel A). It includes 535 and 373 vehicles at the stations from Branches 1 and 2, respectively, and 6,629 driver-agent interactions in the street from Branch 3 (online Appendix Table A2, panel B).

B. The Three Descriptive Facts

Fact 1: The *Real* Revenue Generated Is Large, Mostly Unofficial, and Hinges on the Quota Scheme

Above and beyond the revenue it generated from fines paid at the bank, the agency generated significant revenue from bribes paid in the street to the agents and, more importantly, from station bribes. This revenue was orders of magnitude larger than its revenue from fines. Table 1, panel A, presents the average monthly revenue generated by the staff associated with the average station by source of revenue, including the unassigned intersections sample. Each month, based on Branches 1 and 2, one station generated \$1,816 and \$1,891 of fine revenue paid at the bank, respectively. Yet Branches 1 and 2 reports imply that bribe revenue (street and station bribes pooled) were \$6,007 and \$7,862; these amounted to 77 percent and 81 percent of the sum of fine and (street and station) bribes, respectively (henceforth, total revenue). Total revenue was, therefore, between 4.2 and 5.2 times the official records from fines. Of the revenue from bribes, according to Branch 1, 58 percent

¹¹ Online Appendix Figure A5, panel A shows that, during the rush hour (after 3 PM), Branch 3 source reports 0.3 hourly recorded traffic jams, while, outside the rush hour, it reports 0.1; thus, the measure of traffic jams is informative. As a placebo, panel B shows that, in contrast, recorded accidents are not more prevalent during the rush hour. Online Appendix Figure A6 validates these data by comparing the consistency across measures of traffic characteristics. In an intersection-hour block with a recorded traffic jam, twice as many vehicles drive through per minute, their guessed speed is half, and traffic is half as likely to be fluid.

TABLE 1—FACT 1: REAL REVENUE IS LARGE, MOSTLY UNOFFICIAL, HINGES ON QUOTA

<i>Panel A. The revenue (\$)</i>		
Total revenue	7,823.05	9,753.11
[Ratio]	[4.3 × Fines]	[5.2 × Fines]
(Source)	(Branch 1+3)	(Branch 2+3)
—Revenue from fines	1,816.37	1,891.48
[Percentage of fine + bribe revenue]	[23%]	[19%]
(Source)	(Branch 1)	(Branch 2)
—Revenue from bribes	6,006.68	7,861.63
[Percentage of fine + bribe revenue]	[77%]	[81%]
(Source)	(Branch 1+3)	(Branch 2+3)
— Of which, station bribes	3,476.95	5,331.90
[Percentage of bribe revenue]	[58%]	[68%]
(Source)	(Branch 1)	(Branch 2)
— Of which, street bribes	2,529.73	2,529.73
[Percentage of bribe revenue]	[42%]	[32%]
(Source)	(Branch 3)	(Branch 3)
<i>Panel B. The quota</i>		
Number of drivers requested		294.23
(Source)		(Quota sources)
Number of drivers escorted to the station	207.28	259.66
(Source)	(Branch 1)	(Branch 2)
—Number of drivers who pay a fine	25.13	19.49
[Mean fine]	[\$45]	[\$50]
(Source)	(Branch 1)	(Branch 2)
—Number of drivers who pay a negotiation bribe	104.22	115.06
[Mean negotiation bribe]	[\$12]	[\$14]
(Source)	(Branch 1)	(Branch 2)
—Number of drivers who pay a pound bribe	26.56	51.86
[Mean pound bribe]	[\$11]	[\$8]
(Source)	(Branch 1)	(Branch 2)

Notes: Revenues and transactions induced by station and street staff of one station (on average, out of the eight) over one month (on average). Sample: Descriptive sample + unassigned intersections sample.

is from station bribes; it is even 68 percent according to Branch 2. The remaining 42 and 32 percent according to Branches 1 and 2, respectively, are from bribes paid to agents in the street. The discrepancy in bribe revenue between Branches 1 and 2 arises from station bribes. As discussed in Section IIIC, Branch 2 reports of station *bribes* are less prone to underreporting. We can thus conclude that at least 81 percent of revenue is from bribes; of that bribe revenue, 68 percent is from station bribes. Overall, a station's total revenue is 5.2 times its fine revenue.

These station bribes, which were mostly charge negotiation bribes, were generated by the quota scheme. Table 1, panel B shows that, over 1 month, 1 police station requested 294 vehicles from its agents, who escorted 260 vehicles; of those escorted, 19 paid a fine, 115 paid a charge negotiation bribe, and 52 paid a car pound bribe, with means of \$50, \$14, and \$8, respectively.

In sum, the revenue *really* generated by the agency, regardless of whether it is official or legal, was obtained in large part through an unofficial scheme sustained by a coalition of officials and enabled by the agency's official organization and was much larger than fine revenue. The magnitude of this *real* revenue is surprisingly large, compared to the official revenue statistics, which can only hinge on the revenue from the fines, for a state with purportedly weak capacity. We now provide

evidence of how *exactly* this scheme generated this revenue, through what happened at the stations, and then how that was enabled by actions of the agents.

Fact 2: The Revenue Generated through the Scheme Hinges, in Part, on Extortion in the Stations

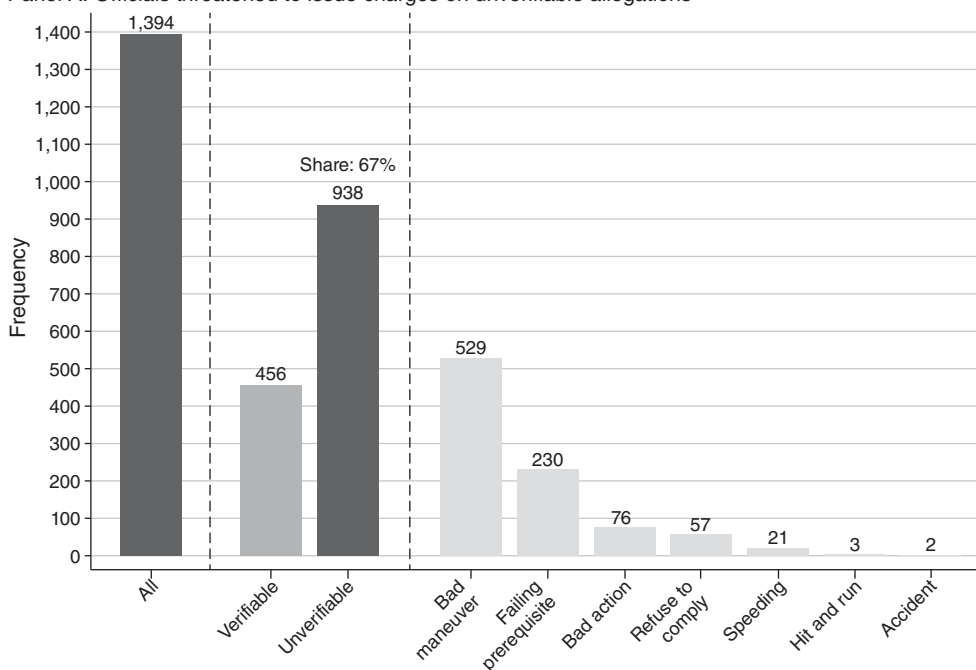
Figure 2, panel A shows that, of all 1,394 negotiations over infractions allegedly made by escorted drivers in Branch 1, 938 (67 percent) were allegations that third parties would be unable to verify (unverifiable). Those relied on the claims made by the agents and included allegations that the driver made a bad or illegal maneuver or that they refused to comply. Qualitative notes suggest that these allegations were sometimes false, which would mean that, colluding with the agents to accept or even encourage false allegations, station officials were able to extort drivers. Furthermore, online Appendix Figure A7, panel A shows that the average station bribe *paid* for an unverifiable allegation was mildly larger than for the rest; thus, in addition to potentially having the discretion to extort drivers, the station officials also had an *incentive* to do so.

What was the source of their power in making such threats? Once the agent escorted the driver to the station, the JPOs threatened the driver with a charge based on the allegation provided by the agent. Using all events recorded in Branch 1's data collection sample, online Appendix Figure A7, panel B shows that 32 percent of escorted drivers call a protector upon receiving the threat of a charge at the station and that protectors were high-ranked police and military officials. Individuals with these positions are known to routinely use the power of the state to exert blackmail (Malukisa and Titeca 2018). Figure 2, panel B, shows the steps in the station bribe negotiation. First, while the mean initial station bribe request was close to \$60 regardless of whether the driver subsequently called a protector, only 52 percent of those who did call one paid a station bribe, against 71 percent for those who did not. Among those who paid one, those who called a protector paid \$14.9, but those who did not call one paid \$24.3. This implies that not having a protector they could call increased the expected station bribe from \$7.8 to \$17.2.¹² Drivers who did not call a protector were three times more likely to be charged. Overall, this indicates that the power of the protectors was to credibly impose a cost on the station officials if they charged the driver.

In sum, this suggests that station bribes, the pillar of the quota scheme, could seemingly sometimes be extortion in the stations, hinging on the station officials' discretion whether to issue charges based on the agent's unverifiable reports. The station officials' ability to do so was thus in part dependent on the discretion of the agents to escort drivers even when they had not made an infraction, and to bring arbitrary allegations for the station officials to negotiate over.

¹²Online Appendix Table A6 shows that the *gap* between the initial bribe asked by the JPO (larger on average) and the actual bribe paid (smaller on average) is larger for drivers with a protector.

Panel A. Officials threatened to issue charges on unverifiable allegations



Panel B. Drivers who called a powerful “protector” paid less station bribes

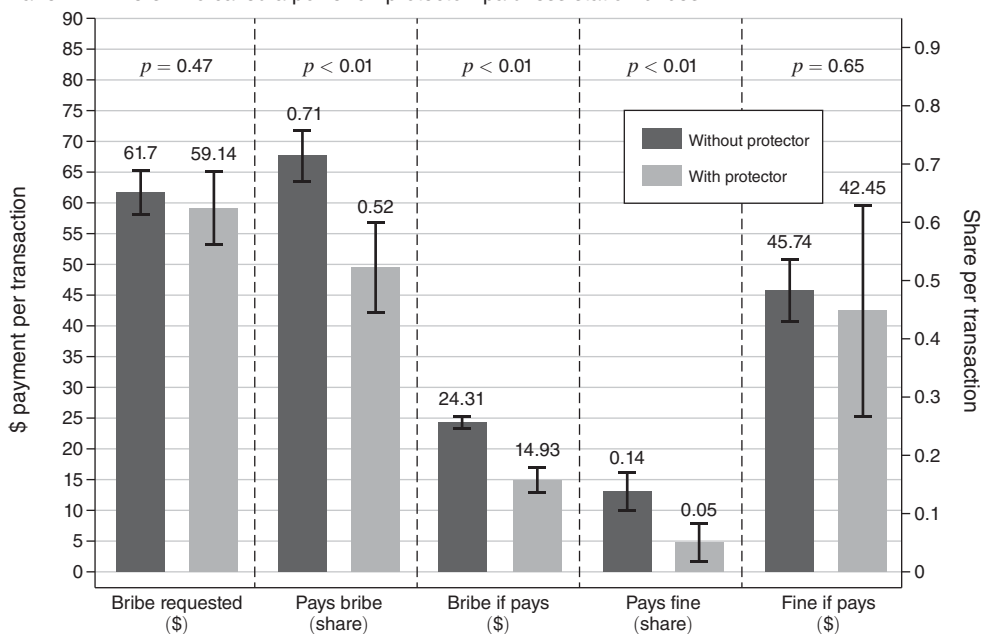


FIGURE 2. FACT 2: THE SCHEME’S REVENUE HINGES, IN PART, ON EXTORTION IN THE STATIONS

Notes: Panel A presents the frequency of alleged infractions for each of the 1,394 reported infractions. Unverifiable allegations have been grouped into simple categories, which include bad or illegal stop, open door, not wearing a helmet, seat belt unfastened, refusing to comply, traffic obstructions, illegally boarded passengers, false maneuver, and traffic lane violation. Panel B documents the mean outcomes in each step of the station bribes negotiation process between the JPOs and the drivers in the station, for drivers who did not call a protector versus drivers who did. p denotes the p -value for a test of equality between means in dark versus light bars. The brackets indicate 95 percent confidence intervals.

Source in both panels: Station events from Branch 1.

TABLE 2—FACT 3: EXTORTION AT THE STATIONS HINGES ON THE AGENTS' POWER TO ARBITRARILY ESCORT DRIVERS AND TO ARBITRARILY ALLEGE INFRACTIONS

	Count	Percentage
<i>Pull-overs</i>	1,841	100
Open-ended description missing	246	13
Open-ended description nonmissing	1,595	87
<i>Panel A. Alleged infraction leading to pull-over</i>		
By claim		
Clearly true	328	18
Not clearly true (dubious)	1,267	69
By driver's response		
Clearly disputed	533	29
Not clearly disputed	1,062	58
By outcome: (Source: hard-coded outcome)		
Driver escorted	1,285	70
to station	417	23
to unknown location	868	47
Driver escaped	179	10
Bribe (bribe) paid	377	20
<i>Panel B. Use of force during pull-over</i>		
Clearly coerced	372	20
Not clearly coerced	1,223	67

Note: This table presents the mean characteristics of traffic pull-overs in the street by the agents.

Source: Unless otherwise noted, the data in this table are from manually coding the open-ended descriptions of each driver-agent interaction typed by Branch 3 observers; where noted, the source is Branch 3 hard-coded data. Sample: Data collection sample. The data collection sample, described in Section IIIA, is constituted of 351 team-days, constituted of 14 teams on Day Zero, and 337 team-days after Day Zero (the days of the data collection sample after Day Zero are the experiment sample). There are 13,793 agent-driver interactions in the data collection sample, of which 1,841 are pull-overs.

Fact 3: Extortion at the Stations Hinges on the Agents' Power to Arbitrarily Escort Drivers and to Arbitrarily Allege Infractions

To examine the nature of allegations made by the agents, we analyze the process of escorting, which hinges on pulling over drivers in the street. Table 2, panel A, shows that of 1,841 events in which an agent pulls over a driver in the street in the data collection sample, Branch 3 observers saw proof that the driver committed an infraction in only 328 (18 percent); e.g., “The rider did not wear a helmet!” “Was unable to show any document,” “was parked in a pedestrian lane.”¹³ The rest were based on no clearly apparent infraction or were simply dubious. Pull-overs were frequently seen as illegitimate by the drivers. Indeed, panel A also shows that drivers escaped 10 percent (181) of the traffic pull-overs, and in 29 percent of them, they engaged in heated disputes with the agents. Yet pull-over legitimacy was intimately connected to whether there was a true infraction. Indeed, online Appendix Table A7,

¹³In contrast, absence of evidence of an infraction does not imply evidence that no infraction was committed, as the driver may have done something wrong they did not see, such as whether the license, if privately seen by the agent, was expired. In some cases, they could say it was clearly arbitrary “*all his documents were in order*,” but it is infrequent.

using the 1,841 pull-overs, shows that pull-overs with clearly true infractions induce more cooperation by the drivers (lower rates of escape, higher rates of a bargained bribe outcome, and almost zero disputes) and induce significantly less brute force by the agents. Taken together, this suggests that the agents routinely pulled over drivers based on claims that were unclear or illegitimate; consistent with existing qualitative literature (Baaz and Olsson 2011), this also supports that they were poorly related to whether the driver violated the code.

What power did the agents use that enabled them to arbitrarily escort drivers? They often used physical force. Table 2, panel B shows that agents used force in at least 20 percent of pull-overs (Qualitative Quotes Appendix (shown in online Appendix, p. 43), Section I.). For example, agents routinely used force to seize the key, the battery, or the steering wheel, pinching the tires or leading to escape. The observers' open-ended descriptions of the escorts' attempts involved threats, brutality, disputes, injury, insults, fights, and bullying.¹⁴ Qualitative evidence suggests that agents routinely weaponized their official discretion to produce allegations in order to arbitrarily escort drivers shielded from repercussions, as well as to support the JPOs' negotiation over station bribes in the station. For example, Branch 2 handwritten notes describe that agents escorted drivers under the pretense of "fantasy infractions":

At the traffic police agency, there is what we call "serfdom," which means that they work for their boss. This is why there are fantasy arrests to satisfy their commander vis-à-vis the quota that he requests. For example: defective brake light, yet when he [the driver] comes to the station, the driver declares, and the JPO verifies, and it is actually working.

(Qualitative notes from Branch 2 observer, July 2015)

This suggests that agents often escorted drivers regardless of whether the driver had complied with the law. And, when false claims might be implausible, they caused infractions themselves: "I really observe that the agents are not honest; they cause infractions" (Street observer end-of-day open-ended comments, July 2015).

In sum, this suggests that the agents regularly used force to pull over drivers and escort them to the stations seemingly arbitrarily, and thus that escorting, the pillar of quota, could be in part under extortion. Taken together, these facts also imply that drivers who did not have a protector, so long as agents were able to detect that in the street, were a profitable target for pull-overs and escorting; thus, the quota could induce the agents to disproportionately attempt to escort drivers who do not have a protector, if the officers were sufficiently sophisticated and able to agree on it.

C. Validity of the Averages from the Descriptive Sample

The descriptive sample is a representative sample of the data collection sample team-days. However, if the experiment induced spillovers, it may not be representative of the potential outcomes in those team-days without encouragement. First, online Appendix Figure A9 presents estimates of intertemporal and

¹⁴ Descriptions of pull-overs contain more anger, fear, and sadness than the rest of the interactions (online Appendix Figure A8).

intrastation spillovers for the variables used in Facts 1, 2, and 3, respectively. We find no evidence of spillovers within the team across days nor to other teams within the same station-day. Second, online Appendix Figure A10 plots pre-study distributions of proxies for the quota and for the station bribe revenue against the estimates from the descriptive sample. The pre-study proxies were ballpark estimates of the means obtained prior to the study through various months of fieldwork with local informants, which we obtained to design the encouragement experiment and determine its compensation. The means of the descriptive sample and those of the pre-study proxies have comparable magnitudes and are positively associated across intersections. Third, in an exit survey after the completion of the study, we collected the agents' self-reported income; the mean, \$14, is almost identical to the estimates obtained in the descriptive sample (online Appendix Table A5), \$326 monthly, amounting to \$12 to \$14 daily. This analysis, therefore, provides reassurance that the street- and station-level averages based on the descriptive sample are valid estimates of the true means in the period of the data collection.

D. Interpretation

In sum, through the station bribes generated by the quota scheme, the agency had a *real*, if unofficial, capacity to raise revenue five times larger than that from fines. Our estimates imply that the mean ratio of the police officers' bribe income (net of the monetary payments they made to their respective superiors) \$464 and \$584 monthly per capita, over their total income was 86 percent and 85 percent for agents and station police staff, respectively; it was 6.7 and 5.8 times their official wage and 5 and 26 times the *actual* teacher income in DRC, respectively, which was \$80 monthly (which was \$69 and \$100, respectively) (Brandt 2013). This revenue was not generated by isolated state officials. It was shaped by the power the state officials' managers had over their agents, shaping a coalition of officials with different powers vested in them by the organization of the state. It also hinged on the relative informal personal power of officials versus citizens, which influenced extortion. Thus, our results suggest that this particular "weak state," in fact, has a quite strong capacity to generate illicit revenue from its citizens. In the following sections, we examine the effect of the scheme on public service.

V. The Effect of the Quota on Public Service: Analysis of the Experiment

In this section, we present an analysis of the effect of the quota on public service.

A. Organizing Framework

The quota scheme can influence both traffic management and deterrence of code violations. First, if it induces the agent to substitute effort away from traffic management and into escorting, it could reduce their effort to manage traffic (Prediction 1). Effort substitution is reasonable because the agents allocate scarce time across tasks. Second, *in addition*, the effect of the quota on the agency's impact on the deterrence depends on whether the drivers escorted due to the quota have made true infractions. Indeed, if the manager has the power to *extort*—that is, to charge drivers even

when the alleged infraction is false—then the quota could induce the agent to target drivers who can be expected to yield a large bribe rather than those who committed an infraction (Prediction 2a); if, through this targeting, the quota induces the production of more false than true allegations, then the quota could reduce the wedge between the implied cost faced by drivers for complying with the code versus not complying, thus undermining the deterrent effect of the agency (Prediction 2b).¹⁵

This simple framework hinges on four assumptions. First, we assumed that the agents could detect which drivers would generate a large station bribe. In reality, agents have some information about what *types* of drivers, based on their visible characteristics, tend to be high-bribe at the station, such as expensive vehicles. More importantly, after stopping a vehicle, they engage in a negotiation commonly called *tracasserie*; agents routinely use this to extract a “signal” from the driver’s bribe potential. For example, the agent can find out if the driver has a powerful protector; the agent also gets an informal sense of the driver’s ability to wield cash. Thus, the assumption that the agents can condition who they target for escorting on the bribe yield is not unreasonable. Given that more than one stop is needed to identify a high-bribe driver, an additional testable implication from this analysis is that a higher quota should lead to more pull-overs, many of which may not result in escorting. Thus, when analyzing the effect of the quota on traffic outcomes, we also examine whether the quota increases pull-overs. Third, because our experiment is not designed to affect the driver choices, we have omitted those. This raises the question of why drivers are unable to bargain with the agents for a bribe to avoid being escorted or why some drivers do not get protectors. In Sections VB and VC, we take as given that drivers *are* regularly escorted and use the experiment to examine both predictions. Then, in Section VI, we present additional evidence to explore why some drivers cannot avoid being escorted and how that may affect the interpretation of the main effect of the quota. Fourth, in reality, there are typically four agents per team. To prevent moral hazard in teams, the managers always place one team leader (called the *Gradés d’Elite* in French), who, in their words, acted as a “spy” for the manager, whose loyalty appeared to not be a concern; thus, with this organizational design, the cost of the unitary actor assumption for a team of agents is low for our purpose.

B. Effect of the Quota on the First Mandate: Traffic Management

We now use the experimentally induced team-day variation in the encouragement for managers to reduce a team quota to examine the effect of the quota on the agency’s service. The encouragement is bundled with a cash transfer to the manager. On the one hand, this ensures that the quota reduction does not induce an income effect for the manager and thus shuts down possible confounds arising from a behavioral response by the managers motivated by seeking substitute sources of income. On the other, this

¹⁵ A similar argument is made in Olken and Barron (2009). The Mathematical Appendix (shown in online Appendix, p. 40) derives these predictions formally. In the canonical model of collusion in organizations (Tirole 1986), a principal owns the output produced by a citizen and wants to incentivize the effort by the citizen (here, drivers’ code compliance). A supervisor (here, the agency) may have *hard information* about whether the citizen provides effort but can conceal it from the principal in exchange for a bribe from the citizen (bribery); they may also have *soft information*, that is, the power to *forge* evidence that the citizen has shirked: *extortion* (Polinsky and Shavell 2001). While bribery may or may not weaken incentives for effort, extortion can destroy them. The model takes this as given and focuses on the supervisory agency’s manager and its agent.

bundling could affect interpretation in various ways. First, the encouragement could have relaxed the managers' time constraint. Were it to be true, this is unlikely to have had any influence, as it was not a binding constraint for the station operations; reports of the managers at the street were rare, and managers were not involved in the negotiations. Those were led by the JPOs, who spent little time with each driver and waited for vehicles most of the day. Second, the encouragement might have affected the behavior of the agents, for example, if they wanted a share of the manager's compensation. This is implausible, as only the managers were informed about the experiment. To the best of our knowledge, no one else became aware. Third, given the fact that a bribe income substitute was given, the expectation that the experiment would end could have influenced the managers' choice of quotas on days not assigned to the encouragement. For example, they could have raised the quotas in anticipation of harder effort in the future. Learning through the fieldwork, conversations with the managers after the study, and a comparison of study and pre-study estimates of the quota and informal income suggest this is unlikely (online Appendix Figure A10 showed them to be similar).

Estimation.—Let i and t index the team/intersection and t day, respectively. Informed by the design (teams are strata and team-days are clusters) and the randomization balance, we estimate the following equations in the experiment sample (henceforth, baseline specification) in two-stage least squares:

$$(1) \quad y_{it} = \alpha + \beta_Q \hat{Q}_{it} + \eta_i + \eta_t + e_{it}$$

$$(2) \quad Q_{it} = a + \beta_T T_{it}^{quota} + u_{it},$$

where $T_{it}^{quota} \in \{0, 1\}$ is an indicator taking value 1 if team-day it was assigned to the *encouragement*, $Q_{it} \in \mathbb{N}$ is the recorded level of the quota, and η_i and η_t are an intersection i (the strata) and a day t fixed effect. Of the 337 team-days of the experiment sample, records of the quota are nonmissing in 326 team-days. The dependent variable y_{it} is an intersection-day level aggregate of hourly traffic outcomes from Branch 3. We construct intersection-day-hour indicators taking value 1 if at least one of the observers posted to a given intersection reported there to be an event in the corresponding intersection-day-hour. For each traffic outcome (traffic jams and accidents), *incidence* is the sum of these intersection-day-hour indicators in an *intersection-day* (this corresponds to the number of times the traffic outcome event occurred in intersection i day t); *occurrence* is an indicator for whether at least one of the intersection-day-hour indicators takes value 1 in the intersection-day (this indicates whether (at least) a traffic jam or an accident was reported at that intersection-day). We then consider robustness to data construction.¹⁶

Parameter β_Q measures the average effect of a one-driver increase in the quota on y_{it} for *compliers*. The assignment and the dependent variable are both at the level of the team-day, but we present standard errors as follows: not clustered, as implied by the assignment procedure; clustered by station-day; and clustered by intersection,

¹⁶Online Appendix Table A8 suggests that attrition is not correlated with potential outcomes.

calculated using the score test for wild bootstrap inference for instrumental variable regressions (Kline and Santos 2012).

Estimates.—Table 3, panel A presents the estimates of the first-stage coefficients (equation (1)). The sample across columns changes according to the corresponding second-stage regression sample in panel B. The coefficient on assignment to the encouragement is large and statistically significant at the 1 percent significance level; thus, the first stage is strong, and the assignment to the encouragement is a valid instrument for the quota level (which, as has been described, is randomized).

Panel B presents the two-stage least squares estimates of β_Q . In columns 1–4 the dependent variables are the measures of traffic accidents' and traffic jams' incidence and occurrence. An increase in the quota for a team increases accidents and traffic jams at the corresponding intersections. These effects are statistically significant at conventional levels of significance.

Statistical Significance: Clustering at the station-day level (175 clusters) or at the team level (18 clusters) does not affect the p -value. The significance of these estimates is not affected by intertemporal or intrastation spillovers. Table 3, panel B presents these estimates controlling for the lagged treatment indicator, T_{it-1}^{quota} , and for an indicator for whether another team under the same manager was treated, T_{-it}^{quota} . The coefficient and significance with such controls are unchanged. Table 3, panel C presents the intention-to-treat (ITT) estimates:

$$(3) \quad y_{it} = \alpha^{ITT} + \beta^{ITT} T_{it}^{quota} + \eta_i^{ITT} + \eta_t^{ITT} + e_{it}^{ITT}.$$

The encouragement is associated with a statistically significant reduction in both accidents and traffic jams. Statistical significance is preserved when calculating the p -value of the two-sided test of whether β^{ITT} is different than zero using randomization inference (Duflo, Glennerster, and Kremer 2007), clustering the standard errors at the station-day or team (wild bootstrap).

Robustness: First, the estimates are preserved when individual teams, days, or blocks of days are excluded one at a time (online Appendix Figure A11). Second, the effect has the same sign, albeit statistically insignificant, and uses Branch 2 handwritten descriptions of the allegation as accident records, $n = 14$ (online Appendix Table A9, column 1). Third, the effect is preserved using an alternative external source for traffic jams, based on observers traveling on public transport minibuses occasionally passing through the experiment sample's intersections (column 2). Fourth, the quota also decreases the number of drivers who pass by the intersection when there is no traffic jam (columns 3 and 4), consistent with more drivers being stopped. Fifth, the effect is preserved using the *maximum* and *minimum* number of observers in an intersection-day who report a traffic jam or traffic to be fluid, respectively (columns 5–8). Sixth, a reduction at $t - 1$ does not affect the number of drivers passing through at t (online Appendix Table A10), and the number of drivers passing per minute across all intersections does not exhibit a trend across the study period, oscillating around 20 (online Appendix Figure A12), suggesting driver responses are unlikely to explain this effect. Finally, the effect on all other outcomes is unaffected controlling for traffic jams' incidence (online Appendix Table A11).

TABLE 3—EFFECT OF THE QUOTA ON THE FIRST MANDATE: TRAFFIC OUTCOMES

	<i>Quota level</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. First-stage estimates</i>								
<i>Encouragement_{it}</i>	−2.535 (0.272)	−2.535 (0.272)	−2.535 (0.272)	−2.535 (0.272)	−2.517 (0.275)	−2.517 (0.275)	−2.527 (0.277)	−2.737 (0.0866)
First-stage <i>F</i> -statistic	86.61	86.61	86.61	86.61	83.96	83.96	83.45	84.18
Dep. var. mean (no encouragement)	6.54	6.54	6.54	6.54	6.55	6.55	6.50	6.96
Observations	321	321	321	321	318	318	315	2,737
<i>Accidents</i>		<i>Traffic jams</i>		<i>Pull-overs</i>	<i>Team size</i>	<i>Time away</i>	<i>Quota met</i>	
<i>Incidence</i>		<i>Occurrence</i>	<i>Incidence</i>	<i>Occurrence</i>	#	[0,1]	Hours	0/1
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
<i>Panel B. Second-stage IV 2SLS estimates (LATE)</i>								
<i>Quota level_{it}</i>	0.063 (0.022)	0.051 (0.016)	0.145 (0.065)	0.034 (0.018)	1.088 (0.371)	0.001 (0.008)	0.336 (0.154)	−0.063 (0.011)
<i>p</i> -value, SE clustered at police station-day	0.01	0.00	0.03	0.05	0.01	0.95	0.02	0.00
<i>p</i> -value, SE clustered at team	0.00	0.00	0.03	0.05	0.00	0.94	0.01	0.00
Dep. var. mean (encouragement)	0.09	0.08	1.38	0.46	8.89	0.86	2.79	0.32
Observations	321	321	321	321	318	318	315	2,737
<i>Controlling for temporal and organizational lags</i>								
<i>Quota level_{it}</i>	0.056 (0.021)	0.045 (0.015)	0.110 (0.060)	0.030 (0.018)	0.972 (0.353)	0.002 (0.008)	0.328 (0.148)	−0.065 (0.011)
Observations	308	308	308	308	305	305	302	2,630
<i>Panel C. Reduced-form estimates (ITT)</i>								
<i>Encouragement_{it}</i>	−0.156 (0.055)	−0.126 (0.038)	−0.349 (0.168)	−0.084 (0.049)	−2.640 (0.932)	−0.001 (0.022)	−1.015 (0.415)	0.175 (0.036)
<i>p</i> -value, randomization inference	0.00	0.00	0.04	0.08	0.00	0.96	0.01	0.00
<i>p</i> -value, SE clustered at police station-day	0.00	0.00	0.04	0.06	0.01	0.96	0.00	0.00
<i>p</i> -value, SE clustered at team	0.01	0.00	0.06	0.08	0.01	0.95	0.00	0.00
Dep. var. mean (no encouragement)	0.25	0.20	1.47	0.51	11.30	0.87	3.45	0.15
Observations	332	332	332	332	328	328	325	2,783

Notes: Panel A: equation (1) estimates; first-stage *F*-statistic is Olea and Pflueger's (2013) effective *F*-statistic of the first stage. Panel B: equation (1) estimates. *p*-value reports the *p*-value of the test of whether β_Q is different from zero. Standard errors clustered at the level of the team are estimated using wild bootstrap score (Kline and Santos 2012); *p*-values using Wang and Zhang (2022) are qualitatively identical. Controlling for spillovers: estimates of equation (1) also include, as controls, the lag of the treatment indicator, T_{it-1}^{quota} , and the neighbor-team treatment status, T_{-it}^{quota} (taking value 1 if at least one team was treated in the same station and day, and 0 otherwise). Panel C: estimates of equation (3), and applies randomization inference to compute the *p*-value following Duflo, Glennerster, and Kremer (2007) and wild bootstrap test when clustered at the team level. Sample: Experiment sample. In all columns, the observations are at the team-day level (denoted *it*, in the text). The number of team clusters is 18; the number of station-day clusters is 175. "Incidence" is the number of hour blocks in the day in which at least one observer reported a given event; "Occurrence" is whether an event took place in the day. "Team size," aggregated per team per day, is constructed using the street observers' information of whether the randomly selected target agent is actually posted at the intersection that day, described in greater detail in study documentation (shown in online Appendix, p. 48), Section III. "Time away," aggregated per team per day, is the gap between the timestamp at which the observed agent is seen interacting with a driver for an escort to the station and the time stamp of the observed agent's next interaction at the intersection, aggregated at the team-day level. "Quota met" is at hourly level; thus, standard errors in column 8 are clustered at the team-day level.

Source: Branch 3.

Economic Significance: The 2SLS coefficients for incidence imply that, for each 15.8/6.9 drivers requested, there is one more accident/traffic jam at the corresponding intersection, respectively, and that shutting down the quota would reduce the daily incidence of accidents and of traffic jams in all study intersections from 3.75 to 0 (100 percent) and from 22.05 to 13.35 (40 percent), respectively.¹⁷ Even if the intersections are a small subset of Kinshasa's network, and thus the quota likely affects only a small set of the potential events in the network, this nonetheless suggests the quota has a large effect on traffic outcomes in central locations where most drivers transit.

Behavioral Channels: Table 3, columns 5–7 show that the quota increased the number of pull-overs and absenteeism of assigned agents at the corresponding street intersections, but the number of assigned agents working (team size) remained unaffected. Taking as given that 87 percent of the 4.2 mean assigned agents per team are working on a 7-hour day (panel C), these estimates imply that the quota would decrease time present in the street from 25.6 to 23.5 agent-hours, a 2.1 agent-hours, 8 percent decrease. If the effect on accidents is entirely channeled through absenteeism, this would mean that the decrease from 0.25 to 0 accidents per day per intersection (100 percent decrease) that would result from shutting down the quota would be caused by 2.1 additional agent-hours in the street (8 percent increase); a linear extrapolation would imply that 1 accident at the intersection per day is being avoided for (at most) each 8.4 additional agent-hours present (33 percent increase). This implied significance of agent presence is larger than existing quantitative estimates in the literature.¹⁸ While it is possible that the proportional effect of agents' presence in this context is unusually high (which can be rationalized by the high baseline presence of agents, the fact the quota affects the intensive margin rather than the extensive margin, and the fact that there is a low baseline level of accidents), the existing quantitative estimates in other settings suggest it may be implausible that absenteeism alone can explain the whole effect of the quota on traffic outcomes. Branch 3 open-ended descriptions confirm that absenteeism can be, in part, a channel, but suggest other factors, such as the frequency with which agents are a hazard to traffic (excerpts are in qualitative appendix, shown in online Appendix, p. 43, Sections II, III, IV). Whichever are the agents' behavioral adjustments that channel the effect of the quota on traffic outcomes, Table 3, column 8

¹⁷These predictions are obtained as follows: $\hat{y}(Q = 0) = \bar{y}(\bar{Q}) - \hat{\beta}_Q \times \bar{Q}$, where $\bar{y}(\bar{Q})$ is the mean of y_{it} without quota reduction encouragement and $\bar{Q}$ is the mean quota without quota reduction encouragement. Since the encouragement is randomly assigned, the sample mean of y_{it} without quota reduction encouragement is an unbiased estimator of the mean of y_{it} without quota reduction encouragement. Then, we use the estimates for $\hat{\beta}_Q = 0.0629$ and 0.144 for accident and traffic jams incidence, respectively, the median quota level, 4, and the mean incidence of accidents and traffic jams without quota reduction encouragement, 0.25 and 1.47, respectively. Then, the predicted accident incidence at an intersection, were the quota to be 0, is lower than 0, 100 percent lower; the predicted traffic jam incidence is 0.89, 40 percent lower. We obtain the numbers at the level of the 15 study intersections by scaling up in a factor of 15. Using the mean quota of 6.3 produces a reduction of 100 percent and 62 percent. This prediction is preserved if the estimating equation includes quadratic quota-level terms instrumented by the level of encouragement.

¹⁸Same calculation as for traffic outcomes. The meta-analysis by Botteghi, Theofilatos, and Macaluso (2017) includes the following estimates: increasing mobile radar enforcement exposure by 100 percent results in a 21 percent reduction in injury accidents on rural roads (Goldenbeld and Schagen 2005), increasing the average check length by 1 hour results in a 5.07 percent increase in monthly speed-related collisions (Li et al. 2017), increasing police enforcement exposure by 1 percent results in a 0.25 percent reduction in crashes (Elvik 2011), increasing conventional speed enforcement exposure by 1 percent results in a 0.07 percent reduction (Elvik 2015).

shows that increasing the quota implies that the quota is more often *unmet* across the hours of the day; this is consistent with the effect of the quota being channeled through changes in agent behavior resulting from spending more hours in the day working toward meeting the quota.

In sum, the quota worsens the agency's performance on its first mandate, traffic management. However, it does not follow that the quota harms public service provided. Its total effect on the value of service depends on its effect on deterring code violations, to which we now turn.

C. *Effect of the Quota on the Second Mandate: Deterrence*

The study was not designed to analyze driver compliance. Indeed, since the experiment induces idiosyncratic changes in the quota's (secretive) level in a context of frequent changes in agent behavior, it would be unlikely for drivers to respond through idiosyncratic changes in compliance.

Instead, we infer the implications for deterrence by analyzing the agency's behavior. Section VA provided two testable implications for whether the quota undermines deterrence about the characteristics of escorted vehicles. Namely, we showed that if the manager's power to issue false charges is large (high power to extort), and if the agents are able to select who they escort based on the drivers' expected station bribe, then a higher quota would disproportionately induce escorts of vehicles with a high expected station bribe—*Prediction 2a*. Furthermore, if this targeting effect holds, then a large power to extort would translate into the quota leading to the production of more false allegations than true ones—*Prediction 2b*, corresponding to *weaker* deterrence.

Effect of the Quota on Types of Drivers Escorted (Prediction 2a).—We begin by empirically examining whether the quota disproportionately induces the escorting of drivers who could be expected to yield a high station bribe (Prediction 2a). First, we draw on the descriptive evidence presented in Section V to identify drivers who could be expected to yield a high station bribe. Section V showed that drivers without a protector yielded significantly larger station bribe revenue. How can agents target those drivers? As introduced in Section VA, the agents can easily learn in the street in the discussion ensuing a pull-over whether the driver has a powerful protector. Table 4, panel A, columns 1–2 present the estimates of equation (1), with dependent variables the number of escorted drivers escorted by team i on day t that do not, and that do, call a protector, respectively. The estimates show that the quota increases the escorts of drivers who do not call a protector; it does not increase escorts of drivers who call a protector. The coefficient of the former is 15 times larger.

Second, we use the characteristics of escorted drivers to predict the station bribe of escorted vehicles, independent of the actual station bribe paid. Columns 3, 4, 5, and 6 use as dependent variables the number of escorted drivers whose *predicted station bribe* is above/below the median station bribe of the no encouragement sample, and those whose *predicted fine* is above/below the corresponding median. To determine *predicted station bribe* and *predicted fine*, we regress actual station bribes on all observed characteristics in Branch 1 and construct a *predicted station bribe* and a *predicted fine* for each driver using these estimates (results are the same using Lasso instead). Columns 3 and 4 show that the quota increases the escorting of

TABLE 4—EFFECT OF THE QUOTA ON THE SECOND MANDATE: TESTABLE IMPLICATIONS FOR DETERRENCE

	Number of escorts by team-day that are ... (Source: Branch 1, JPOs)						Allegations (Prediction 2b)	
	Targeting of drivers who may yield a high bribe (Prediction 2a)						(Prediction 2b)	
	<i>Unprotected</i>	<i>Protected</i>	<i>High-Bribe</i>	<i>Low-Bribe</i>	<i>High-Fine</i>	<i>Low-Fine</i>	<i>Unverifiable</i>	<i>Verifiable</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. In the stations (Predictions 2a, 2b)</i>								
<i>A. Second-stage IV 2SLS estimates (LATE)</i>								
<i>Quota level_{it}</i>	0.191 (0.0771)	0.0127 (0.0541)	0.159 (0.0712)	0.0486 (0.0588)	0.0768 (0.0849)	0.131 (0.0677)	0.502 (0.123)	0.169 (0.106)
<i>p</i> -value, SE clustered at police station-day	0.01	0.79	0.02	0.42	0.31	0.04	0.00	0.09
<i>p</i> -value, SE clustered at team	0.02	0.79	0.02	0.38	0.30	0.04	0.00	0.08
Dep. var. mean (encouragement)	2.72	1.48	2.53	2.13	2.67	1.99	3.29	1.75
Observations	259	259	236	236	236	236	225	225
First-stage <i>F</i> -statistic	89.66	89.66	101.47	101.47	101.47	101.47	78.42	78.42
Controlling for temporal and organizational lags								
<i>Quota level_{it}</i>	0.180 (0.073)	0.023 (0.051)	0.175 (0.069)	0.039 (0.057)	0.087 (0.081)	0.128 (0.066)	0.485 (0.118)	0.153 (0.103)
Observations	248	248	228	228	228	228	219	219
<i>B. Reduced-form estimates (ITT)</i>								
<i>Encouragement_{it}</i>	−0.526 (0.231)	0.009 (0.166)	−0.436 (0.239)	−0.132 (0.194)	−0.182 (0.260)	−0.385 (0.220)	−1.350 (0.366)	−0.461 (0.318)
<i>p</i> -value, randomization inference	0.02	0.96	0.07	0.50	0.49	0.07	0.00	0.14
<i>p</i> -value, SE clustered at police station-day	0.01	0.96	0.07	0.50	0.49	0.05	0.00	0.12
<i>p</i> -value, SE clustered at team	0.02	0.97	0.06	0.49	0.50	0.06	0.00	0.11
Dep. var. mean (no encouragement)	3.22	1.35	3.04	1.93	3.01	1.95	4.48	2.11
Observations	264	264	241	241	241	241	230	230

(continued)

Notes: Sample: Experiment sample. Of all 326 team-days in the experiment sample for which the quota level is nonmissing, 259 have nonmissing protector or type of vehicle data, and 225 have nonmissing allegations data. In all columns, the observations are the number of vehicles seen escorted at the station from the intersection (team) i on day t with the characteristic listed in the respective table headers, except for columns 7 and 8, which are the number of allegations in the day from the corresponding intersection. There are 17 team clusters and 159 police station-day clusters. Variable construction: The dependent variables High-Bribe, Low-Bribe, High-Fine, and Low-Fine are constructed as follows. After computing the median bribe from team-days not assigned to the encouragement, we estimate, using OLS, the predicted station bribe of an escorted vehicle, using as regressors all the recorded characteristics of escorted drivers reported in Branch 1. Using the estimates from this model, we construct a “station bribe potential score” for each escorted driver and compute, for each team-day, the number of drivers whose station bribe potential score is above the median of station bribes that we computed among escorts from team-days not assigned to the encouragement.

Sources: For the dependent variables, Branch 1 (JPOs); for the endogenous regressor (quota level), as described in Section IIIC.

drivers who, based on this prediction, can be predicted to yield a large station bribe; it does not increase escorts of drivers with low predicted station bribe. Columns 5 and 6 show that the quota induces escorts of drivers with low, rather than high, predicted *fine*; thus, the increase in high-bribe escorts is not an incidental outcome of the quota inducing targeting of high-fine drivers who may incidentally yield high station bribes.

TABLE 4—EFFECT OF THE QUOTA ON THE SECOND MANDATE: TESTABLE IMPLICATIONS FOR DETERRENCE
(continued)

	Infractions and pull-overs			Disapproval	Driver cooperation			Coercion
	Number of pull overs, by infraction			Any event	Outcome of traffic pull over			Any event
	Any	Not clear	True	Dispute	Escape	Escort	Bribe	Force
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel B. In the street (Prediction 2b)</i>								
<i>A. Second-stage IV 2SLS estimates (LATE)</i>								
<i>Quota level_{it}</i>	1.088 (0.371)	0.864 (0.356)	0.211 (0.152)	0.399 (0.197)	0.0269 (0.0135)	0.0252 (0.0146)	−0.0522 (0.0197)	0.282 (0.164)
<i>p</i> -value, SE clustered at police station-day	0.01	0.03	0.17	0.03	0.05	0.08	0.01	0.07
<i>p</i> -value, SE clustered at team	0.00	0.02	0.17	0.03	0.00	0.02	0.00	0.06
Dep. var. mean (encouragement)	8.89	7.02	1.65	2.56	0.06	0.20	0.74	1.75
Observations	318	311	311	318	1,701	1,700	1,701	318
First-stage <i>F</i> -statistic	83.96	77.72	77.72	83.96	26.80	27.09	26.80	83.96
<i>Controlling for temporal and organizational lags</i>								
<i>Quota level_{it}</i>	0.972 (0.353)	0.726 (0.338)	0.215 (0.143)	0.391 (0.192)	0.017 (0.011)	0.025 (0.014)	−0.042 (0.017)	0.269 (0.161)
Observations	305	298	298	305	1,617	1,617	1,617	305
<i>B. Reduced-form estimates (ITT)</i>								
<i>Encouragement_{it}</i>	−2.640 (0.932)	−2.070 (0.889)	−0.540 (0.394)	−0.963 (0.521)	−0.042 (0.024)	−0.048 (0.026)	0.090 (0.031)	−0.675 (0.437)
<i>p</i> -value, randomization inference	0.00	0.01	0.16	0.06	0.00	0.02	0.00	0.10
<i>p</i> -value, SE clustered at police station-day	0.01	0.03	0.17	0.03	0.06	0.06	0.00	0.08
<i>p</i> -value, SE clustered at team	0.01	0.03	0.19	0.03	0.01	0.02	0.00	0.10
Dep. var. mean (no encouragement)	11.30	8.86	2.19	3.68	0.12	0.27	0.61	2.52
Observations	328	321	321	328	1,756	1,753	1,756	328

Notes: Sample: Experiment sample. In columns 1–3 and 7–8, the dependent variables are constructed using the open-ended answers of all events in the experiment sample where open-ended descriptions were missing in this aggregation. Columns 4–6 restrict the analysis to the sample of traffic pull-overs. In columns 1–4 and 8, observations are aggregated at the team-day level; in columns 5–7, observations are at the level of the agent-driver pull-over. Variable construction: Among 1,841 traffic pull-overs, 1,595 had nonmissing open-ended questions and thus could be used to construct the dependent variables in columns 2, 3, 4, and 8; among 1,595 nonmissing open-ended pull-overs, 328 were reported by Branch 3 observers as being based on certainly true claims. Column 7 pools observed bribe payment (*amende* in French) with observed escort to an unknown location, which is often to find a discreet place to pay an *amende*. In columns 1–3, 4, and 8, the dependent variable is a count. In columns 5–7, the dependent variable is an indicator taking value one if the pull-over ends in any of the following outcomes, respectively: escape, escort to the station, direct evidence of bribe, or escort to hidden location (where often a bribe is paid). There are 18 team clusters and 174 police station-day clusters.

Sources: The source for the dependent variables in all columns is Branch 3 street observers; the sources for the endogenous regressor (quota level) are described in Section III.

In sum, the quota fuels *exclusively* escorting of drivers with high predicted station bribe but not with high predicted fine. This is consistent with Prediction 2a, which is a testable implication of the quota causing extortion.

Effect of the Quota on the Type of Allegations (Prediction 2b).—We now turn to analyzing whether the quota scheme induces the agency to produce more false alleged infractions than it induces the production of true ones (Prediction 2b).

In the stations, the quota increased escorts with unverifiable allegations. Using Branch 1 reports, Table 4, panel A, column 7 shows that the quota increases the number of escorted drivers with unverifiable infractions, but it does not increase escorts of drivers with verifiable allegations. This result is robust to intertemporal and intrastation spillovers and to alternative standard error calculations.¹⁹ In sum, this suggests that the quota disproportionately induces the escorting of drivers with allegations that are more likely to be false. It is a testable implication from the quota inducing extortion, but it also establishes that the quota reduces deterrence.

In the street, the quota increased pull-overs where the driver did not clearly make an infraction and that were perceived as illegitimate by the drivers. To establish this result, we proceed in two steps, analyzing Branch 3 reports of agent-driver interactions from the intersections. First, we ask whether the quota induces *more* attempts by the agents to escort drivers on the basis of arbitrary claims, instead of clearly true ones. Table 4, panel B, columns 1–3 show that the quota increases traffic pull-overs, driven by traffic pull-overs based on claims that are not clearly true and hence potentially arbitrary. To examine whether the analysis of an allegation’s truth is meaningful, we undertook manual linking of the street events to the police station events using all of the available vehicle characteristics. Online Appendix Figure A13 shows that vehicles reported by Branch 3 observers as having made a true infraction and escorted to the stations are slightly more likely to have made a verifiable allegation, providing confidence in Branch 3’s assessment. They are also just as likely to pay a bribe or a fine at the station. Overall, this suggests that the quota increases the agents’ arbitrary attempts to escort rather than drivers who clearly committed infractions.

Second, we ask whether the quota induces traffic pull-overs perceived to be illegitimate by the drivers, and thus, which may be false or arbitrary. Recall that we saw in online Appendix Table A7 that when the driver clearly made a true infraction, the driver was less likely to escape a pull-over attempt or to engage in a dispute, and the agent was less likely to use force. Table 4, panel B, column 4 shows that the quota increases the frequency of disputes among agent-driver interactions. Columns 5–7 examine the reallocation of the pull-over outcome, conditional on pull-over events: the driver is more likely to escape or to be escorted; in contrast, the only outcome in which driver-agent bargaining does not “break down,” *amende*, is less likely. This effect could be rationalized by the quota strengthening the agents’ incentive to convert attempts into escorts. However, this simple demand-side explanation would tend to reduce escape, which is the opposite of what column 5 showed and thus inconsistent with this interpretation. Column 8 shows that the quota increases the use of force by the agents, although the effect is marginally significant. Supporting that breakdown of bargaining indicates lower cooperation, a Congolese analyst reported: “Alternatively, the driver has not committed any violation, and is upset due to the mistreatment of the officer. He then agrees to be taken to the office ...

¹⁹ We consider two sources of nonclassical measurement error that may affect the validity of our interpretation. First, the quota could change the conditions in which Branch 1 observers (JPOs) report driver characteristics. Online Appendix Table A12 shows that the quota has no effect on JPO attendance, suggesting nonclassical measurement error may not drive this result. It is also unlikely: as described in Section I, the JPOs are idle at the station most of the day. Second, online Appendix Table A13, panel A shows that these results are preserved by controlling for traffic jams’ incidence, suggesting that, as before, endogenous visibility due to congestion does not explain these results.

Sometimes, the driver demands that he be given a payment slip to pay the fine at the bank and give nothing to the police” (interview with Congolese analyst, 2023).

Taken together, these results underscore that the quota induces effort by the agents to pull over drivers on the basis of arbitrary criteria for which it is implausible that true infractions were made—extortion.²⁰

In sum, guided by the framework in Section VA, this section has provided evidence suggesting that the quota also undermines deterrence, the agency’s second mandate. Together with its effect on traffic outcomes, this suggests that, compared to corruption without the scheme, the scheme subverts public service, a social cost above and beyond that from agent corruption.

VI. Why the Vehicle Quota? Drivers, Managers, and Agents’ Incentive

In this section, we examine whether the quota scheme can be rationalized as an equilibrium corruption contract in this setting. First, we examine why agents and escorted drivers could not bargain over a bribe to avoid escorting. As we have seen, this can happen, leading to the payment of an *amende* (bribe payment related to bargaining over not being escorted); yet it fails for escorted drivers. Were bargaining to always succeed, the vehicle quota could not be rational. Our qualitative evidence suggests that the vehicle quota hinges on this bargaining often breaking down for three reasons. One reason for its breakdown is simply liquidity constraints; the drivers usually have little cash on hand: “I’ll give an example ... The driver ... has only 30 dollars, but the police refuse this money. They demand 50 dollars, and the driver, in turn, makes it clear to them that he only has that amount. Dissatisfied, the police take him to the office where he is asked to pay 100 dollars to be released” (interview with project manager, 2023).

To support this qualitative observation, we use data from Lameke et al. (2024). These data show that even minibus drivers, who made cash through the day, had limited liquidity: accounting for their revenue from passenger rides and operational costs (gas and repairs), the typical minibus would have \$14.5 at hand on average at midday and less in the hours before.

A second reason is that, in the station, it was often the driver’s protector or the vehicle’s owner, not the driver, who would have to come in order to negotiate and pay on behalf of the driver; the agent was often agent for a third party: “The driver agrees to go to the office because he has the option to call an officer who will allow him to pay no fine, or to pay less” (interview with Congolese analyst, 2023).

A third reason is that the driver or agent occasionally develops spite toward another as a result of the human frictions in the nature of this negotiation:

“At the time of the arrest, the driver angered the police with his behavior (insults, threats to call an officer, attempting to hit an officer, etc.).”

²⁰ All estimates in this panel are robust to spillovers and standard error calculation. Online Appendix Table A13, panel B shows that these results are preserved controlling for traffic jams’ incidence.

These are transaction costs. Their regularity can explain why agents and drivers often fail to bargain a monetary payment, and thus, why the quota does not unravel into monetary kickbacks.

Second, we examine whether the quota can be interpreted as a corruption contract. An alternative interpretation of these results is that the quota may instead be a fiscal tool imposed by the hierarchy of the state on the stations in order to generate fines and that, incidentally, this creates opportunities for station bribes to be collected. Table 5 presents the estimates of equation (2) from the experiment sample. The dependent variable in columns 1 and 2 is measures of fine revenue generated by the vehicles escorted by each team-day to their respective station. They show that the quota has no effect on fine revenue generated at the station from the escorts provided by the team. In contrast, columns 3 and 4, which show the estimates for the dependent variable being the sum of station bribe revenues generated in the day by the vehicles escorted by the corresponding team, show that the quota significantly increases station bribes. Taken together, columns 1–4 are consistent with the quota being a corruption contract rather than a fiscal instrument that creates opportunities for corruption at the stations.

Third, column 5 presents the estimates when the dependent variable is, instead, the revenue by the team-day from all types of street bribes (toll fee, tips, and *amende*). The quota does not reduce the team's revenue from street bribes. This is not surprising. Indeed, it is consistent with the common strategy, which is also rational, of having an agent escort the driver to the police stations while keeping the other in the intersection collecting street bribes. Thus, the technology of street bribe collection, manifested in the weak relationship between the number of agents in the street and street bribes, is a pillar of the quota's optimality. Where this relationship may be steeper, a trade-off might render a vehicle quota less obvious. Overall, this suggests that the quota at its current level maximizes corruption joint profits compared to a lower quota.

In sum, the quota appears to be a rational response by a coalition of agents, station managers, and judicial police officers to the specific institutional details of this setting. It appears to reflect joint profit-maximization motive from the perspective of this coalition, and its optimality for this group hinges on frequent transaction costs between the agents and the drivers, the power of station officials to collect station bribes, and on the technology of street bribe collection.

VII. Corruption Systems in Perspective

In this section, we discuss how specific our findings are, using a qualitative description of similar schemes across the agency, the country, and across the globe and in history.

A. Corruption Systems within the Agency

The interviews we conducted with officials reveal that Kinshasa's traffic police agency did not always have a vehicle quota. According to them, since 1975, their wages have become insufficient for survival. While bribe taking purportedly followed the reduction in their wages, it only began to be organized as a scheme in the 1980s (Nkuku 2022). Then, officers began making payments between \$10 and \$15

TABLE 5—THE QUOTA AS A JOINT CORRUPTION PROFIT-MAXIMIZATION CONTRACT

	Fine revenue (\$)		Bribe revenue (\$)		
	Police station		Police station		Street
	Branch 1 (1)	Branch 2 (2)	Branch 1 (3)	Branch 2 (4)	Branch 3 (5)
<i>Panel A. Second-stage IV 2SLS estimates (LATE)</i>					
<i>Quota level_{it}</i>	2.246 (1.488)	−0.317 (1.421)	3.386 (1.937)	3.259 (1.943)	0.401 (1.906)
<i>p</i> -value, SE clustered at <i>police station-day</i>	0.14	0.84	0.10	0.08	0.83
<i>p</i> -value, SE clustered at <i>team</i>	0.18	0.84	0.10	0.09	0.82
Dep. var. mean (encouragement)	18.64	18.82	59.28	62.88	51.25
Observations	259	173	259	173	318
First-stage <i>F</i> -statistic	89.66	73.27	89.66	89.66	83.96
<i>Controlling for temporal and organizational lags</i>					
<i>Quota level_{it}</i>	1.943 (1.425)	−0.612 (1.475)	3.437 (1.816)	3.329 (1.991)	0.673 (1.866)
Observations	248	172	248	172	305
<i>Panel B. Reduced-form estimates (ITT)</i>					
<i>Encouragement_{it}</i>	−6.013 (4.514)	1.032 (5.130)	−8.639 (5.852)	−10.606 (6.938)	−0.511 (5.022)
<i>p</i> -value, randomization inference	0.18	0.85	0.12	0.14	0.91
<i>p</i> -value, SE clustered at <i>police station-day</i>	0.20	0.86	0.16	0.10	0.92
<i>p</i> -value, SE clustered at <i>team</i>	0.24	0.85	0.19	0.12	0.91
Dep. var. mean (no encouragement)	22.69	19.03	66.94	70.31	53.30
Observations	264	173	264	173	328

Notes: Sample: Experiment sample. The experiment sample covers 337 team-day observations, which are also the units of assignment, 326 of which have nonmissing quota level. Of all 337 team-days in the experiment sample, 264 (Branch 1) and 173 (Branch 2) have nonmissing station revenue data, and 248 (Branch 1) and 173 (Branch 2) have both nonmissing quota level and escort revenue data; 328 have nonmissing street bribe revenue data, and 318 have both nonmissing quota level and street bribe revenue data. In all columns, the observations are at the team-day level (denoted *it*, in the text) and refer to the revenues from the vehicles seen escorted at the station from the intersection (hence, team) *i* on day *t* with the characteristic listed in the respective table headers. There are 17 team clusters and 157 station clusters.

Sources: Branches 1, 2, and 3, described in Section IIIB. Branch 3 events are aggregated at the team-day level following the aggregation procedure detailed in online Study Documentation, (shown in online Appendix, p. 43), Section III; the sources for the endogenous regressor (quota level) are described in Section IIIB.

dollars daily to their commanders, who then made payments to higher levels of the hierarchy, purportedly under the oversight of a provincial commissioner; this practice came to be known as “retrocession.”

After 2010, the demands of this practice on the officers increased. Commanders were asked to make weekly payments to a newly created structure, the *Coordination* in French (Titeca and Malukisa 2014), and these payments amounted to a significant share of the bribes they were able to collect themselves directly. Members of the “Coordination” then purportedly made payments to a Minister and a national commissioner. Some agents referred to that scheme as a “commercial enterprise.”²¹

²¹“Through the Coordination, the traffic police became a commercial enterprise; but only for [two commissioners]!” (interview with agent, 2013).

Agents considered asking money from drivers to be legitimate but described the payment requirements to their superiors as “servitude” or being a “blood donor” (Titeca and Malukisa 2014). Sanctions for not paying also increased (see qualitative Appendix (shown in online Appendix, p. 43), Section V).

Our interviews suggest that, up to 2014, the practice of paying the superiors changed into a quota of vehicles, reportedly in three steps. First, between 2010 and 2014, the number of stations increased from 2 to 29; thus, intersections got closer to any station. The commanders at the stations began requesting vehicles rather than money. According to one commander, this practice allowed to preserve a “facade of legality”: it followed the official procedure of bringing the vehicle to the station, which protected the agents from being arrested for taking bribes in the street. Second, in late 2014, as President Kabila had become unpopular in the capital, a new general of the Kinshasa police introduced measures to increase the president’s popularity. Taking bribes in the street became severely penalized.²² The qualitative interviews we conducted during this time suggest that bribes paid by drivers in the street decreased and that the vehicle quota scheme became entrenched. Third, taking street bribes eventually stopped being risky; even then, the vehicle quota scheme practice continued: commanders had reportedly learned through this experience that it was more profitable than relying on “retrocession.” One commander said, “Before, the agents could exaggerate in these negotiations and obtain a lot of money, of which they only gave a small amount to the commander. Now we, or our responsible agents, can directly negotiate with the drivers. It always is better to directly negotiate with the drivers” (interview with commander, 2013).

Across the agency’s locations, the schemes varied. In Lumumbashi, the colonels obtained weekly payments from the agents in cash (so-called “reports”). In Goma, the scheme amounted to a vehicle quota scheme, but agents were allowed to substitute vehicles for cash at a given exchange rate (qualitative interviews in Goma and Lubumbashi; see qualitative Appendix (shown in online Appendix, p. 43), Section V).

In sum, the way in which bribe-taking schemes were organized in the traffic police agency varied over the years in Kinsahasa and across cities. Its specific organization appeared to reflect a combination of rationality, habits, and experimentation.

B. Corruption Systems in Global Context

This scheme could be understood as an exceptional characteristic of a weak state with little relevance to other settings. Indeed, the DRC has been described as a “paradigm of informalisation and criminalisation of the state” (Petit and Mutambwa 2005), created over 30 years of Mobutu rule. Ex-President Mobutu’s famous quote “*Moto na moto abongisa*” or “Let each person sort things out at his own level” sent a strong signal to state officials generally; it was “ironically” reinterpreted in relation to bribe taking and rule breaking, as an invitation for “each and everyone to steal at his own level of responsibility” (Petit and Mutambwa 2005, p. 482). As a

²²The general frequently posed undercover in the street, leading to arrests of various officers. The agents referred to the climate as the “Kanyama psychosis” and to the general as “*Esprit de Mort*” (spirit of death) due to his hard line.

result, state positions have been described as becoming “privatized” (Lemarchand 1988): officials were encouraged to “steal cleverly” (Titeca and De Herdt 2011). Although the public revenues eventually increased after Mobutu’s departure in 1997 due to international aid throughout the 2000s (Titeca and De Herdt 2011), the practices created during Mobutu’s rule remained. Titeca and De Herdt (2011) document such systems in the education sector; Verweijen (2013) in the military; and Lake (2022) in other branches of the police (see qualitative Appendix (shown in online Appendix, p. 43), Section V).

However, similar schemes were documented across the world. Because of the illicit and organized nature of the corruption schemes involving managers and their agents, academic research on such schemes is rather limited. Among academic research, Wade (1982) concluded that organized corruption “may be a more fundamental influence on Indian administration and politics than has hitherto been realised,” for which Pant (1999) provided evidence in a different state. Weaver (2021), in a remarkable study, shows evidence of bribes for jobs in a public agency.

Notwithstanding the limited academic research on this topic, public legal investigations reveal that these schemes can be frequent and large. In middle- and low-income countries, there is evidence of such schemes within public firms and in the education and health sectors. First, regarding state-owned firms, as recently as 2014, one of the largest corruption scandals in history was uncovered in Brazil. Operation Car Wash (*Operação Lava Jato*) revealed a money-laundering and bribery scheme involving Petrobras, Brazil’s state-controlled oil company. Reportedly, executives at Petrobras overpaid suppliers and persuaded them to channel back to the officials a share of the overpaying, who then shared some of it in kickbacks to politicians who gave them the position (Londono 2019; Watts 2017; *Economist* 2022). Second, regarding the education sector, recently, evidence suggested corruption schemes in financing systems in the South Africa Department of Higher Education (Simango 2024): “We believe [the conspiracy] is carried out by sophisticated and protected syndicates ... linked to the upper echelons of the Department of Higher Education.” Third, recent studies suggest this might be prevalent in the health sector. While these studies report nepotism and cronyism across the board, sometimes with smoking gun evidence of rent-sharing agreements that incentivize lower-level corruption, the latter has been explicitly documented in some cases. For example, in Uganda, “promotion was dependent on continued demonstration of loyalty to the appointing patron by participating in other corrupt acts such as demanding informal payments (bribery) and embezzlement” (Kiryia 2020). In Greece, for example, evidence became abundant after the investigations that followed the outbreak of the financial crisis, pointing to grand corruption schemes. Schemes were reportedly found involving the operation of a major hospital, the compensations for diagnostic arthroscopies, the pharmaceutical policy and drug supplies, a health entity involved in clientelism and patronage, and large pharmaceutical companies. In general, it was reported that health suppliers paid kickbacks to influence the referral and prescription process of doctors and the procurement process in health structures (Stamouli, Gasparinatos, and Kouroutzas 2023).

In the United States, there is also evidence of similar schemes. For example, concerning police forces, the Knapp Commission report revealed in the 1970s that “a large part of [New York] city’s police force has been guilty of some form

of illegal activity,” where “both high-ranking police officers and high-ranking members of the Lindsay administration had Information about the scope of this activity but did nothing about it,” suggesting that in order to incentivize looking the other way, rent-sharing agreements might have occurred (*New York Times* 1973). Transnational corruption schemes have also been documented in the health sector. Indeed, recently, an investigation reported evidence supporting a possible grand corruption scheme in global supply chains in “the largest ever project funded by the US Agency for International Development (USAID)” within its first two years of its operation (Stockton, Igoe, and Khan 2023): “Inadequate financial monitoring ... combined with potential collusion between Chemonics [International, a private international development firm based in Washington, DC] and Zenith [subcontractor of Chemonics International] staff, allowed the fraud to remain undetected.” The investigation “found that an unnamed senior project director allegedly implicated in the collusion, who signed off on all distribution invoices, was ‘living substantially beyond their Chemonics salary’,” pointing toward a scheme.

Evidence from investigations thus paints a picture of manager-agent schemes inside state administrations as potentially relevant across the globe but also potentially sophisticated and large. Yet the evidence from investigated cases likely represents only the “tip of the iceberg” of all existing corruption schemes; it is limited to those that are discovered and investigated.

Historical records provide evidence that the practice was also common throughout history across the globe. The public administrations in France, Spain, Italy, the Netherlands, Germany, the Ottoman Empire, and China, as well as the colonial administration of European empires are replete with similar examples (Swart 1949). The practices in that period (e.g., *venalité des offices* in France’s Ancien Régime) developed out of the practice of farming out offices, which was widespread in the Middle Ages, with the idea that public office was a source of profit rather than a duty toward a king. In France, as early as the fourteenth century, notaries and tax collectors were allowed to transfer their offices for a payment (Dupont-Ferrier 1930). In the next century, it was prevalent among sheriffs and courts of justice. After the end of the fifteenth century, it spread to provide cash for rulers in wars (Ardant 1975).²³

In sum, Kinshasa’s traffic police scheme would appear to be a rather common consequence of the personal privileges provided by a state administration rather than an exception.

VIII. Conclusion

The results of this study suggest that when institutions are underfunded and cannot pay the wage of their officials, state officials and their managers may use the official powers bestowed upon them to collude against the public, shaping corruption schemes that support real revenue-generating capacity and that may be even more distortionary than corruption without a scheme. This has some implications for policy. First, interventions aiming to curb corruption could focus on the levels of the hierarchy above the frontline civil servants and on the ways in which the official

²³ Francois I even created an agency in charge of all the sales of offices, named the *Bureau des Parties Casuelles* (Dupont-Ferrier 1640), and the annual sum to be paid, called *la Paulette*, began to be centrally dictated.

powers bestowed upon officials at different levels can create opportunities for collusion against the public. This could also have implications for policing more generally (e.g., Ba et al. 2021; Rivera and Ba 2023). Second, in the context of enforcement, technologies that reduce the officials' discretion to fabricate allegations could mitigate the effect of corruption arising specifically from extortion—dampening the incentives for citizens to comply with the law—even when corruption continues to exist; furthermore, the efficacy of such technology hinges on the specific organizational design of the state, e.g., who has the discretion to make allegations or to issue charges and what is the relationship between them.

Our study suggests that corruption may be the “tip of an iceberg” involving broader coalitions of state officials and suggests three possible avenues for research. First, the scheme is just one small cog in a broader system. Agents, drivers, and commanders regularly make large payments to higher-ups to seek their “protection.” In qualitative interviews, we found that station managers made monthly unofficial payments to individuals higher up in the hierarchy, amounting on average to \$225.53 monthly. Understanding the players, strategies, games played, what they maximize, and how equilibria structure the behavior of its participants is a promising avenue of future research. Second, this context may be prevalent in many weak states, but, even as we have found evidence of their existence across settings, it is likely that they are either less prevalent or present in a different form in high-income countries and in low-income countries with stronger states. Given the challenges of uncovering such schemes, more research is needed to even determine how prevalent they are beyond the public scandals for which we have provided evidence. Third, we do not understand how such schemes are “embedded” in specific cultural ways of understanding relationships (e.g., the concept of *Guanxi* networks in China (Yin 2017)). The power of managers may also matter in understanding corruption schemes beyond their material power: their role as leaders may also shape organizational cultures of corruption; e.g., the “behavior of superiors” is often reported to be the most influential factor on unethical decisions (Baumhart 1961; Brenner and Molander 1977).

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